Inequality in a Potential Equalizer of Opportunity*

Variation in the Accessibility and Net Benefits of English-Medium Instruction in India

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Abstract

English-medium instruction (EMI), or the teaching of school subjects in English, is often viewed as a potential palliative for the intergenerational rigidity of socioeconomic status in India, where English skills command substantial wage premia. I study whether the conditions governing households' access to and potential net gains from EMI are themselves unequally distributed. I organize my analysis around three such conditions: inherited linguistic costs to acquiring English, access to English-medium schooling, and the local economic setting in which English skills may be rewarded. By harmonizing district data on languages, schooling, consumption, migration, human capital, and sectoral employment, I find inequality across all three. Inherited linguistic conditions are highly persistent and predict future human capital gains, financial asset access, and structural change, but their relationship with language acquisition behavior varies across institutional settings. EMI exposure and provision are strongly organized by schooling system and sharply stratified along socioeconomic lines, even within the same local education market. Consumption growth is more associated with EMI enrollment than enrollment alone, and the strength of their association increases in pre-existing human capital and urbanization. In sum, a high-return educational opportunity need not equalize opportunity when access to it and the resources which help benefit from it are unequally distributed.

1 Introduction

In an Indian economy characterized by rapid GDP growth alongside wage stagnation, profuse informality, and high rates of both youth unemployment and underemployment (International Labour Organization 2024), what can a household do to improve the future prospects of their children? The history of economics highlights *education* as one answer (Mincer 1974; Card 1999; Björklund and Kjellström 2002; Bhandari and Bordoloi 2006). The history of India highlights another: *English*. In a context of extraordinary linguistic diversity (Gurevich et al. 2025) and

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thus extraordinary communication and coordination costs (Nakagawa and Sugasawa 2021), English has come to serve as a primary domestic lingua franca (Clingingsmith 2014). In its capacity as a global lingua franca (Melitz 2016), meanwhile, English opens pathways both internationally—e.g., to education at prestigious universities or employment at multinational firms (Ramanathan 2014; Bedi 2019, 32–33)—and at home. The domestic expansion of internationally traded services and information technology, in particular, has led to a flurry of jobs whose higher socioeconomic returns (Chamarbagwala and Sharma 2011; Smokotin et al. 2014) are largely accessible only to those who can use English (Mankiw and Swagel 2006; Shastry 2012).

Empirical study corroborates the notion that English expands economic mobility (Itani et al. 2015). Using nationally representative household data, Azam et al. (2013) find that English fluency was, at least up until 2005, associated with a 34% hourly wage premium for men and a 22% premium for women; even limited English proficiency was associated with premia of 13% and 10%, respectively. Refeque and Azad (2022) find similar differentials extending through 2011–12. Evidence for causality in this association comes from Chakraborty and Bakshi (2016), who interpret West Bengal's 1983–2004 ban on English teaching in public primary schools as a natural experiment where cohorts more *exposed* to the policy earned lower subsequent wages. "Language skills are human capital," after all (Chiswick and Miller 1995, 4), and in a context like India especially, "English-language skills [specifically] are a form of human capital" (Azam et al. 2013, 344).

It is thus by combining our two answers—education and English—that we can begin to understand the demand for *English-medium instruction (EMI)*, or for the teaching of all school subjects in English, currently seen in India (Gooptu and Mukherjee 2023; Lahoti and Mukhopadhyay 2019; Chattopadhay and Roy 2017). For its potential to expand equality of opportunity, EMI has been touted as a tool of social development (National Knowledge Commission 2009) and described as a route to escape caste (Mathur 2013; Mathew and Srivastava 2011; Vij 2020) alongside the rigid wage, employment, and occupation distributions it enforces across generations (Madheswaran and Singhari 2016; Munshi and Rosenzweig 2006).

But it is ultimately households who decide whether or not a child attends EMI, and the determinants of their choice—the costs they face, the benefits they expect, and the forces constraining the choice set in front of them—are affected by factors which are not just national and historical, but also specific and local. My focus is on whether such factors are associated with the same inequalities that English skills are often expected to help households escape. I study inequalities which exist along dimensions such as social group, religion, parental education, household resources, and urban or rural residence. If the ability to access or benefit from EMI is systematically related to these characteristics, even within the same local education market, then we have evidence that a high economic payoff to English skills alone is not enough for EMI to become an equalizer of opportunity.

Though I never estimate a structural model of household choice, their investment problem still provides a helpful

organizing framework. I will first study a type of cost whose variation is predetermined relative to the institutional and economic factors that otherwise affect the household's problem: **inherited linguistic costs to learning English**. These will reflect historically persistent features of the local linguistic environment (see Rowe and Weisleder 2020) which affect the opportunity cost of learning English over other domestic bridge languages, namely Hindi. This work, detailed in Section 3, owes much credit to Shastry (2012); in line with her methodology, I create several different measures of local linguistic distance from *Hindi* to capture this opportunity cost. In Section 4 I analyze the **accessibility of English-medium instruction** through enrollment, various measures of EMI exposure, and private school management. Finally, in Section 5 I study potential determinants of **expected returns to English skills**. My goal here is to find evidence on the value of English skills in the local labor market, with the assumption that households incorporate the local history of such returns into their beliefs regarding English, akin to households having adaptive expectations. I ask whether EMI exposure predicts later district consumption and whether that association is stronger where pre-existing human capital and urbanization are greater. I then examine whether inherited linguistic conditions accompany human-capital growth and skilled in-migration, as well as changes in financial access and economic structure.

The local linguistic environment, the local education market, and the local labor market—all three of the factors I focus on are studied locally. These choices are justifiable: the linguistic conditions which households pass down to their children are highly persistent at the local level, local schooling conditions constrain the menu of educational opportunities households choose from, and local economic conditions plausibly cause households to update their beliefs on the economic returns to learning English. I will use a household's district to proxy for their 'local' environment.[1] Though the district is the most granular unit at which I can harmonize the longitudinal data sources I rely on, it must be noted that interdistrict migration would confound and attenuate any association between the characteristics of one district's education market and its future economic outcomes. Older studies found interdistrict migration rates (and particularly rates of economically-motivated interdistrict migration) to be so low even after the expansion of IT and services in the 1990s (Topalova 2005; National Statistical Office 2021) that they "assume[d] that district of current residence (or of employment) of an individual is approximately the same as the schooling district" (Chakraborty and Bakshi 2016, 6). More recent work, however, complicates (Mahapatro 2020; Rajan et al. 2020) and even outright repudiates (Lucas 2021) this notion. These studies emphasize the role of the IT expansion in driving interdistrict migration (Economic Divison 2015; Ghose 2026), and my own results associate potential determinants of household investment into EMI with the skill composition of recent work migrants. I return to those results in Section B. Regardless, it is evident that district outcomes are not an aggregation of individual returns to EMI, but a function of places and their changing populations.

[1] A district in India is a political unit akin to a county in the United States, albeit with an average population of around 2 million. Table 1 contains summary statistics derived from the distribution of district populations.

The results show that the conditions surrounding English-skill investment are themselves unequally distributed. The relative cost of acquiring a bridge language is partly inherited, and language behavior varies with these inherited linguistic conditions. Formal English-medium schooling is strongly organized by state boundaries, particularly in the public sector. Within the same district, social-group differences remain large in English-medium and private-school participation, with larger differences in urban schooling markets. EMI exposure is more strongly associated with later local consumption than enrollment alone, and this association is stronger in districts with higher initial human capital and urbanization. The inherited linguistic gradient is also associated with later changes in human capital, the skill composition of recent work migrants, financial access, and sectoral composition.

Taken together, the evidence limits the equalizing promise of a high-return educational opportunity. Households begin from unequal linguistic conditions and face unequal access to EMI; among districts, the later consumption association is stronger where baseline human capital and urbanization are higher. English skills can be valuable while access to English-medium schooling and the local conditions associated with its later economic returns remain unequal.

2 Data and Variables

I construct the district datasets used in the paper by reconciling changing district boundaries, reconstructing archived schooling records, linking Census mother tongues to several linguistic classifications, and harmonizing household consumption surveys across major changes in survey design. I place survey, Census, and school records from different boundary regimes onto 2001 Census district geography using a reviewed district-lineage system informed by official identifiers, district histories, and published concordances (Kumar and Somanathan 2016; Ministry of Panchayati Raj, Government of India 2026; Asher et al. 2021). I reconstruct district-level EMI and school-system measures from archived DISE workbooks and report cards whose formats and medium-of-instruction fields vary across releases (Mehta 2022; District Information System for Education 2017). I combine Census mother-tongue populations with Shastry's Hindi-distance classification, reviewed Census-language crosswalks, Glottolog genealogy, and Dyen lexical data (Shastry 2012; Office of the Registrar General and Census Commissioner, India 2004; Hammarström et al. 2026; Dyen et al. 1992; "Comparative Indo-European Database" 1997). I construct comparable district consumption measures from nine NSS/HCES survey designs spanning 2000–01 through 2023–24 (National Sample Survey Organisation 2002, 2003), placing households on 2001 Census district geography and expressing consumption in 2011–12 prices. Five survey designs provide the baseline and later outcomes used in the main longitudinal analysis (National Sample Survey Office 2007, 2012b, 2014, 2024b, 2025a). Construction details appear in Section A, including the additional Census and Economic Census panels

used for local development outcomes. Identification checks are reported in Section D.

Geography and coverage. District boundaries change often enough that a district name cannot serve as a stable longitudinal identifier. I use 2001 Census districts as the reference geography and reconcile later and earlier units using reviewed district-lineage evidence. The lineage system draws on Census materials and independent district-harmonization resources, including the India State and District Evolution Database, the Indian Census Data Collection, the India District Changes Tracker, and SHRUG (Jolad et al. 2026; Jolad and Singh 2026; Jaacks et al. 2020; Asher et al. 2021). The reference geography contains 593 2001 Census districts. The 2007–08 schooling survey reaches 587 of them under the reviewed lineage, while the primary 2017–18 lineage reaches 578; requiring support in both schooling rounds yields a 573-district panel. Because the household surveys are repeated cross-sections rather than repeated observations of the same respondents, I aggregate the relevant measures to harmonized districts.

The principal schooling measures come from the 2007–08 National Sample Survey (NSS) 64th Round, *Participation and Expenditure in Education* (National Sample Survey Office 2011). I use survey weights when forming district aggregates. DISE provides a separate school-reported measure of medium of instruction, management, enrollment, and school characteristics. I reconstruct the DISE district series from report cards covering 2005–06 through 2007–08 that were preserved online after the original publication pages disappeared (Mehta 2022; District Information System for Education 2017). 2001 Census language tables provide mother-tongue populations and language behavior (Office of the Registrar General and Census Commissioner of India 2001). Later sections also use 2001 Census and 2011 Census migration, household, housing, and worker tables. I use the 2005 and 2013–14 Economic Censuses for changes in local economic structure (Central Statistics Office 2008, 2016). Labor outcomes come from the 2009–10 NSS employment survey and the 2017–18 Periodic Labour Force Survey (National Sample Survey Office 2010; National Statistical Office 2019).

Access to English-medium instruction. The NSS schooling variables separate school participation from medium of instruction. Let N_d be the survey-weighted number of children ages 5–19 in district d, E_d the number enrolled, K_d the number enrolled for whom medium of instruction is observed, and M_d the number in K_d enrolled in English-medium instruction (EMI). I call

$$\text{Enrollment}_d = 100\frac{E_d}{N_d} \tag{1}$$

the *enrollment rate*. This is the extensive margin: whether a child is enrolled at all. Among enrolled children

whose medium of instruction is observed, I call

$$\text{EnglishMediumShare}_d = 100\frac{M_d}{K_d} \tag{2}$$

the *English-medium share among enrolled children*. This is the conditional medium-of-instruction margin among children with a reported medium. Finally,

$$\text{EnglishMediumExposure}_d = 100\frac{M_d}{N_d} \tag{3}$$

is *English-medium exposure among all children*. It incorporates enrollment, whether medium is reported among enrolled children, and EMI conditional on a reported medium. Enrolled children with unreported medium remain in an explicit unclassified category. EMI exposure among all children is therefore lower than the conditional English-medium share whenever some children are unenrolled or some enrolled children have unreported medium. I also distinguish *public-school English-medium exposure* and *private-school English-medium exposure*, each defined as the share of all children observed in English-medium schooling in the relevant management sector. Children whose medium or school management is unreported remain outside those sector-medium cells. *Private-school enrollment* is the share of enrolled children with observed school management who attend private aided or unaided schools. These measures describe observed formal-school enrollment and reported medium of instruction. NSS 64 excludes classes taken from private tutors and therefore does not measure supplemental English instruction (National Sample Survey Office 2011).

Inherited linguistic conditions. The primary language measure begins with Shastry's 0–5 ordinal scale of separation from Hindi, developed with linguist Jay Jasanoff (Shastry 2012). Shastry uses linguistic distance from Hindi to capture variation in the relative cost of acquiring English rather than Hindi as a lingua franca. For each mapped, non-English mother tongue l in district d, let S_{ld} denote its 2001 Census speaker population and D_l its Shastry distance from Hindi. The *speaker-weighted distance from Hindi* is

$$\overline{D}_d = \frac{\sum_l S_{ld} D_l \, \mathbf{1}\{D_l > 0\}}{\sum_l S_{ld} \, \mathbf{1}\{D_l > 0\}}. \tag{4}$$

The positive-distance restriction excludes Hindi and Urdu, the zero-distance reference languages, from both the numerator and denominator. The resulting mean describes the linguistic composition of speakers whose mapped mother tongue has positive distance from Hindi. I use *linguistic distance from Hindi* as a shorter name for this conditional speaker-weighted measure in the main text. A second measure, the *share speaking languages distant from Hindi*, is the percentage of all mother-tongue speakers in the district whose mapped language has Shastry

distance at least three. The distant-language share uses all mother-tongue speakers in its denominator, so the size of the distant-language population enters the measure directly.

I use several alternatives when testing whether results depend on one linguistic classification. The *top-three-language distance from Hindi* is the speaker-weighted Shastry distance among the three largest mapped mother tongues in a district, corresponding to the older construction used in the original analysis. *Genealogical distance from Hindi* is the unweighted number of ancestry-tree edges separating each language from Hindi in Glottolog (Hammarström et al. 2026). *Lexical noncognacy with Hindi* is one hundred minus the percentage of cognate items shared with Hindi in the Dyen lexicostatistical data (Dyen et al. 1992; "Comparative Indo-European Database" 1997). I also reconstruct the speaker-weighted and distant-language measures using the 1991 Language Atlas for historical validation (Office of the Registrar General and Census Commissioner, India 2004). The crosswalk and adjudication rules for Census mother tongues are documented in Section A.

Language-behavior controls. The Census language-acquisition regressions use state-by-native-language cells rather than districts, so their controls differ from the district covariates used elsewhere. Within each state I condition on the native language's share of speakers and an indicator for the state's most common native language. The relevant continuous and threshold distance specifications also include an indicator for Hindi/Urdu, the zero-distance reference category. I refer to the first two variables together as *within-state language-prevalence controls*. Their summary statistics appear separately in Table 1 because their unit of observation is a state-language cell rather than a district.

Economic outcomes and baseline covariates. I use household consumption as the principal measure of local material living standards. At the household level, monthly per-capita consumption expenditure (MPCE) is consumption expenditure divided by household size. I express MPCE in 2011–12 prices and estimate district means using the survey design and person weights, so the district statistic is the person-level average of real MPCE implied by the survey weights. The baseline comes from the 2004–05 NSS 61st Round (National Sample Survey Office 2007). The 2009–10 and 2011–12 NSS rounds provide intermediate endpoints (National Sample Survey Office 2012b, 2014), and HCES 2022–23 and 2023–24 provide the long-run endpoints (National Sample Survey Office 2024b, 2025a). I reconstruct each survey's household consumption measure under its documented design and express the resulting MPCE series in 2011–12 prices using official price series, including the Reserve Bank of India and Labour Bureau series (Reserve Bank of India 2026; Labour Bureau, Government of India 2026a). HCES 2022–23 and 2023–24 divide consumption across three questionnaires administered to each selected household in three consecutive monthly visits (National Sample Survey Office 2024b, 2025a). I reconstruct household MPCE from those questionnaire components before district estimation; Section A gives the details.

The preferred district regressions compare districts within states and adjust for covariates measured in 2001,

before the 2007–08 schooling outcomes. I call these the *baseline 2001 Census covariates*. These covariates cover six groups: scale and geography, social composition, human capital, demography, economic structure, and basic development. They include log population, urban share, and log population density; Scheduled Caste, Scheduled Tribe, and Muslim shares; secondary-or-higher attainment; the dependency ratio; the agricultural share of workers; and household electricity access (Office of the Registrar General and Census Commissioner of India 2001; Asher et al. 2021). These covariates adjust the district regressions for observed differences in baseline conditions. In the heterogeneity analysis, baseline secondary-or-higher attainment and urban share enter as moderators and are omitted from the baseline covariate vector.

Table 1 reports the scale and dispersion of the principal variables and both covariate families. Enrollment averages about 74 percent of children ages 5–19, while English-medium exposure among all children averages about 15 percent. Private-school English-medium exposure averages about 9 percent and public-school English-medium exposure about 6 percent. The speaker-weighted distance from Hindi averages 3.05 Shastry degrees across districts, and the district mean of the share speaking languages at distance three or greater is 42 percent. The table also reports all ten baseline district covariates and, in a separate state-language panel, the variables used in the Census language-behavior regressions. Mean real consumption in the available district samples is about Rs. 928 per month in 2004–05, Rs. 2,152 in 2022–23, and Rs. 2,242 in 2023–24, all expressed in 2011–12 prices.

Table 1: Core Variables and Summary Statistics

Variable	Mean	SD	N	p_{10}	p_{90}	Year / unit
Panel A. Principal variables:						
Enrollment	74.44	9.72	573	62.07	87.12	2007-08; % of children age 5-19
English-medium share among enrolled	18.99	25.34	573	0.83	52.98	2007-08; % of enrolled children
English-medium exposure among all children	14.88	20.33	573	0.53	43.89	2007-08; % of children age 5-19
Public-school English-medium exposure	5.55	14.26	573	0.00	15.56	2007-08; % of children age 5-19
Private-school English-medium exposure	9.20	11.58	573	0.32	25.41	2007-08; % of children age 5-19
Private-school enrollment	27.11	19.40	573	3.23	55.43	2007-08; % of enrolled children with known management
Speaker-weighted distance from Hindi	3.05	1.36	573	1.04	4.97	2001; Shastry degrees
Share speaking languages distant from Hindi	42.23	42.80	573	0.13	98.44	2001; % of mother-tongue speakers
Real mean consumption per person	928.07	377.75	557	464.61	1419.84	2004-05; 2011-12-price Rs/person/month
Real mean consumption per person	2152.01	618.67	550	1490.97	2952.44	2022-23; 2011-12-price Rs/person/month
Real mean consumption per person	2242.07	601.13	551	1622.74	2991.47	2023-24; 2011-12-price Rs/person/month
Panel B. Baseline 2001 Census covariates:						
Log population	14.00	1.02	573	12.64	15.03	2001; log persons
Urban population share	23.27	19.11	573	6.18	47.54	2001; % of population
Log population density	5.77	1.16	573	4.57	6.90	2001; log persons/km^2
Scheduled Caste share	14.71	8.64	573	1.96	25.09	2001; % of population
Scheduled Tribe share	16.16	25.85	573	0.00	56.59	2001; % of population
Muslim share	11.40	14.81	573	0.82	24.24	2001; % of population
Secondary-plus share, age 7+	17.09	7.63	573	8.48	27.84	2001; % of population age 7+

Table 1: Core Variables and Summary Statistics *(continued)*

Variable	Mean	SD	N	p_{10}	p_{90}	Year / unit
Dependency ratio	69.39	13.29	573	51.72	86.88	2001; dependents per 100 persons age 15-64
Agricultural worker share	60.53	20.40	573	31.36	81.24	2001; % of workers
Electricity access share	54.44	28.01	573	14.78	89.61	2001; % of households
Panel C. Language-behavior regression controls:						
Native-language share within state	1.26	8.35	2784	0.00	0.50	2001; % of state native-language speakers
State modal-language indicator	0.01	0.11	2784	0.00	0.00	2001; indicator
Hindi/Urdu reference indicator	0.03	0.16	2784	0.00	0.00	2001; indicator

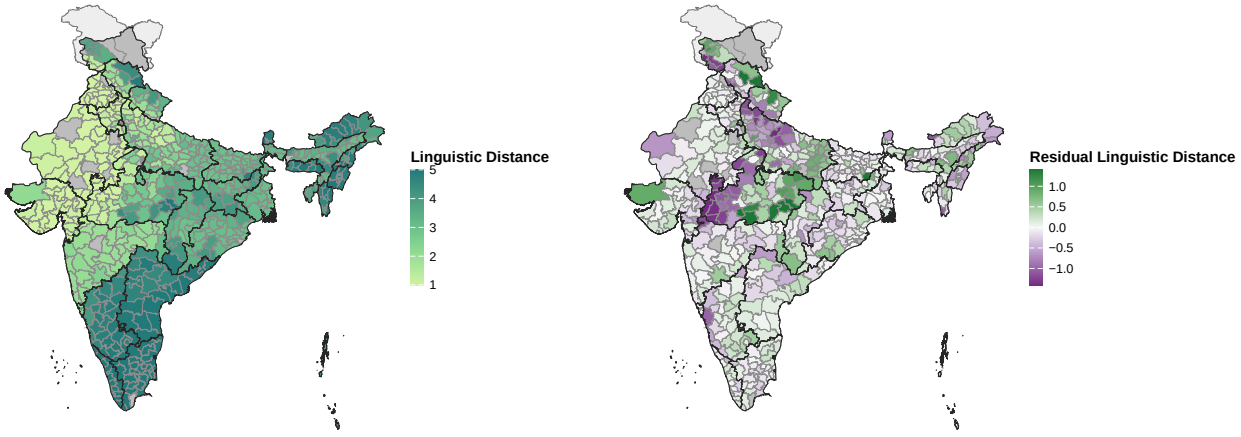
Note: District rows are unweighted descriptive statistics; N is the number of districts with nonmissing values. Schooling rows summarize NSS 2007-08 children ages 5-19. Linguistic distance and the ten baseline district covariates are aligned to 2001 Census districts. The language-behavior controls are summarized separately across 2001 Census state-by-native-language cells because those regressions use a different unit of observation; native-language share is displayed in percentage points. Modern consumption rows use the district samples retained for the corresponding consumption regressions. p_{10} and p_{90} are the 10th and 90th percentiles. Sources: NSS 64th Round, 2007–08; Census of India 2001; NSS 61st Round, 2004–05; HCES 2022–23; HCES 2023–24.

3 Inherited Linguistic Conditions

3.1 Proxies and Persistence of Linguistic Conditions

Why measure distance from Hindi? India's linguistic diversity gives languages of wider communication an important role, and both Hindi and English have served that role across groups (Clingingsmith 2014; Shastry 2012). Shastry's argument is comparative: linguistic diversity creates incentives to learn English or Hindi, and greater distance between a mother tongue and Hindi raises the cost of learning Hindi relative to English (Shastry 2012). The speaker-weighted measure defined in Section 2 summarizes a persistent linguistic component of this relative cost.

Most district variation in the measure occurs between states. In the 573-district schooling panel used later in the paper, state indicators alone explain 84.4% of its cross-district variation. The preferred district specifications therefore compare districts within states and adjust for the baseline 2001 Census covariates. Figure 1 shows the observed measure and the residual after removing state indicators and the linear contribution of those covariates.



(a) Raw linguistic distance.

(b) Residual linguistic distance after state fixed effects and baseline 2001 Census covariates.

Figure 1: District linguistic distance before and after adjustment for state fixed effects and baseline 2001 Census covariates. Sources: Census of India 2001 language and district characteristics (Office of the Registrar General and Census Commissioner of India 2001; Asher et al. 2021). Tengshe (2014) and Natural Earth (2022) provide geospatial data. District boundaries reflect 2001 geography. Politically disputed regions are identified using Natural Earth, and those without data—Aksai Chin, Azad Kashmir, Gilgit-Baltistan, the Trans-Karakoram Tract, and Siachen Glacier—are colored light gray. Their inclusion does not imply a position on territorial disputes.

I call these conditions *inherited* because the district pattern is highly persistent across the independently reconstructed 1991 and 2001 measures. Among the 91 2001 Census districts meeting the historical reconstruction

requirements in both years, the correlation between the 1991 and 2001 speaker-weighted distance measures is 0.898; weighting districts by their 1991 populations raises the correlation to 0.928. The population-weighted correlation for the share of speakers whose mother tongues are at least three Shastry degrees from Hindi is 0.919. Figure 2 shows the district comparison. These estimates agree with Shastry's finding that district linguistic distance is highly correlated across earlier Census years (Shastry 2012).

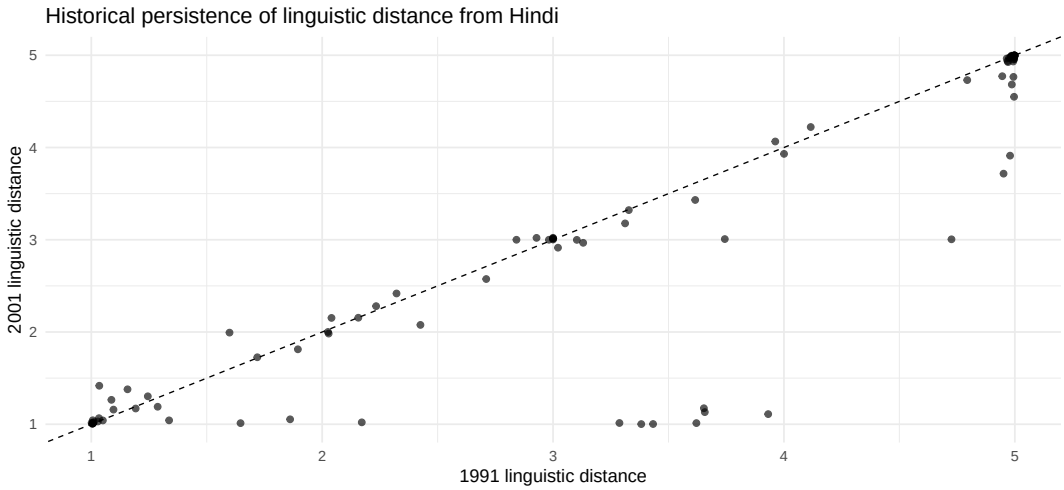


Figure 2: Persistence of speaker-weighted linguistic distance from Hindi from 1991 to 2001. Each point is a 2001 Census district with a reconstructed measure in both years; the dashed line is the 45-degree line. Sources: Census of India 1991 Language Atlas (Office of the Registrar General and Census Commissioner, India 2004) and Census of India 2001 language tables (Office of the Registrar General and Census Commissioner of India 2001).

An independent reconstruction provides a second validation of the 1991 series. Helms and Lim's replication data include a Shastry-derived 1991 district linguistic-distance measure (Helms and Lim 2025). Across the 148 districts shared by their extract and my reconstruction, the two measures correlate at 0.989. The agreement shows that the independently assembled series contain nearly identical cross-district variation.

The measure captures inherited linguistic distance from Hindi. Family members, peers, schools, media, and other social settings also shape the language input children receive (Rowe and Weisleder 2020).

3.2 Linguistic Conditions and Reported Language Choice

I next test whether the Shastry relative-cost measure is associated with the subsidiary languages people report. The 2001 Census bilingualism and trilingualism tables record native-language populations and reported subsidiary languages (Office of the Registrar General and Census Commissioner of India 2001). For English and Hindi, the outcome is the share of multilingual native speakers who report the language as a first subsidiary language or, where applicable, a second subsidiary language. The multilingual outcome is the share of native speakers who

report any subsidiary language. Each regression observation is a state-by-native-language cell, weighted by its native-speaker population. The specifications compare language groups within states and include the within-state language-prevalence controls defined in Section 2; the continuous and threshold specifications also include the Hindi/Urdu reference indicator.

Table 2

Linguistic Distance and Reported Language Choice

	All states				Hindi-belt states		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Linguistic distance	2.234		-9.694***	-3.581***	-0.037		-9.024*
	(1.448)		(2.573)	(0.916)	(2.942)		(5.321)
Distant-language indicator		16.727***				10.165	
		(4.394)				(10.807)	
Outcome	Reported English	Reported English	Reported Hindi	Multilingualism	Reported English	Reported English	Reported Hindi
Population	Multilingual speakers	Multilingual speakers	Multilingual speakers	Native speakers	Multilingual speakers	Multilingual speakers	Multilingual speakers
Observations	1,566	1,566	1,566	1,587	560	560	560
Partial R^2	0.025	0.128	0.094	0.139	0.000	0.046	0.121
State fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Language-prevalence controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Each column is a weighted state-by-native-language regression. Standard errors in parentheses use HC1 heteroskedasticity-robust inference. All specifications include state fixed effects, native-language state share, an indicator for the state's most common native language, and a Hindi/Urdu reference indicator. Continuous-distance coefficients are percentage-point changes per one Shastry distance degree; the distant-language indicator equals one for languages at least three degrees from Hindi. Partial R^2 reports the incremental explanatory power of the displayed linguistic-distance term. Columns (5)-(7) use the prespecified Hindi-belt sample. The coefficients describe associations with reported subsidiary-language outcomes. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Source: Census of India 2001 bilingualism and trilingualism tables.

The continuous English estimate is 2.23 percentage points per Shastry distance degree, with a standard error of 1.45 ($p = 0.123$). The threshold specification is much larger: speakers of languages at least three degrees from Hindi report English at a rate 16.73 percentage points higher than speakers of the remaining classified mother tongues (SE 4.39; $p < 0.001$). This specification parallels Shastry's district measure based on the share of speakers whose languages are at least three degrees from Hindi (Shastry 2012).

The largest continuous coefficient in the table is for reported Hindi. Each additional Shastry degree is associated with 9.69 percentage points less reported Hindi among multilingual speakers (SE 2.57; $p < 0.001$). Greater distance is also associated with a 3.58 percentage-point decline in the share of native speakers who report any subsidiary language (SE 0.92; $p < 0.001$). The pattern therefore includes both the reported subsidiary language and the prevalence of multilingualism.

The alternative linguistic measures produce different English estimates. Glottolog genealogical distance yields a coefficient of 0.21 percentage points (SE 0.43; $p = 0.627$), and Dyen lexical noncognacy yields 0.11 percentage points (SE 0.08; $p = 0.167$) in the corresponding continuous specifications (Hammarström et al. 2026; Dyen et al. 1992; "Comparative Indo-European Database" 1997). The Shastry threshold specification yields the large English association.

3.3 Context Dependence of Reported Language Choice

Within the predeclared Hindi-belt sample, the continuous English coefficient is −0.04 percentage points per Shastry degree (SE 2.94; $p = 0.990$). The threshold estimate is 10.16 percentage points (SE 10.81; $p = 0.347$). Reported Hindi remains negatively associated with distance at 9.02 percentage points per degree (SE 5.32; $p = 0.091$).

The 1991 Scheduled Tribe language data show a different allocation across reported languages (Office of the Registrar General and Census Commissioner, India 2004). In the validated all-state sample, each Shastry degree is associated with 3.38 percentage points more reported Hindi and 0.11 percentage points less reported English. The corresponding bilingualism estimate is 3.64 percentage points. In the Hindi-belt restriction, the Hindi estimate is 5.69 percentage points per degree. The English estimate is −0.17 percentage points, and the bilingualism estimate is 5.82 percentage points. That restriction contains four state clusters, so I use it as descriptive evidence. Clingingsmith's historical analysis provides independent evidence that changing economic incentives increased second-language learning in India (Clingingsmith 2014).

The sex-specific estimates are similar. In the 2001 Census language data (Office of the Registrar General and Census Commissioner of India 2001), the continuous English coefficients are 2.33 percentage points for men and 2.01 percentage points for women per Shastry distance degree. Geographic and historical comparisons produce

much larger differences in the relationship between linguistic distance and reported language.

Across these samples, linguistic distance from Hindi captures persistent variation in the relative cost of learning Hindi and English. The reported subsidiary language associated with greater distance differs across settings. In the national 2001 estimates, greater distance is associated with less Hindi, while the threshold specification shows more English. The English relationship is weak within the Hindi belt. In the 1991 Scheduled Tribe data, greater distance is associated primarily with more Hindi. Reported bridge language choice therefore varies with setting as well as inherited linguistic conditions. The next section examines formal schooling as one institutional setting in which access to English differs.

4 Access to English-Medium Instruction

4.1 Measuring Access to English-Medium Schooling

Inherited linguistic conditions can change the relative cost of acquiring English, but households can invest only through schooling options they can reach. I therefore use the measures defined in Section 2 to describe realized access to English-medium schooling. The NSS 64th Round records household choices in 2007–08: whether children ages 5–19 are enrolled, the reported medium of instruction among enrolled children, and school management among enrolled children with reported management (National Sample Survey Office 2011). These outcomes jointly reflect available schools, household preferences, and household resources.

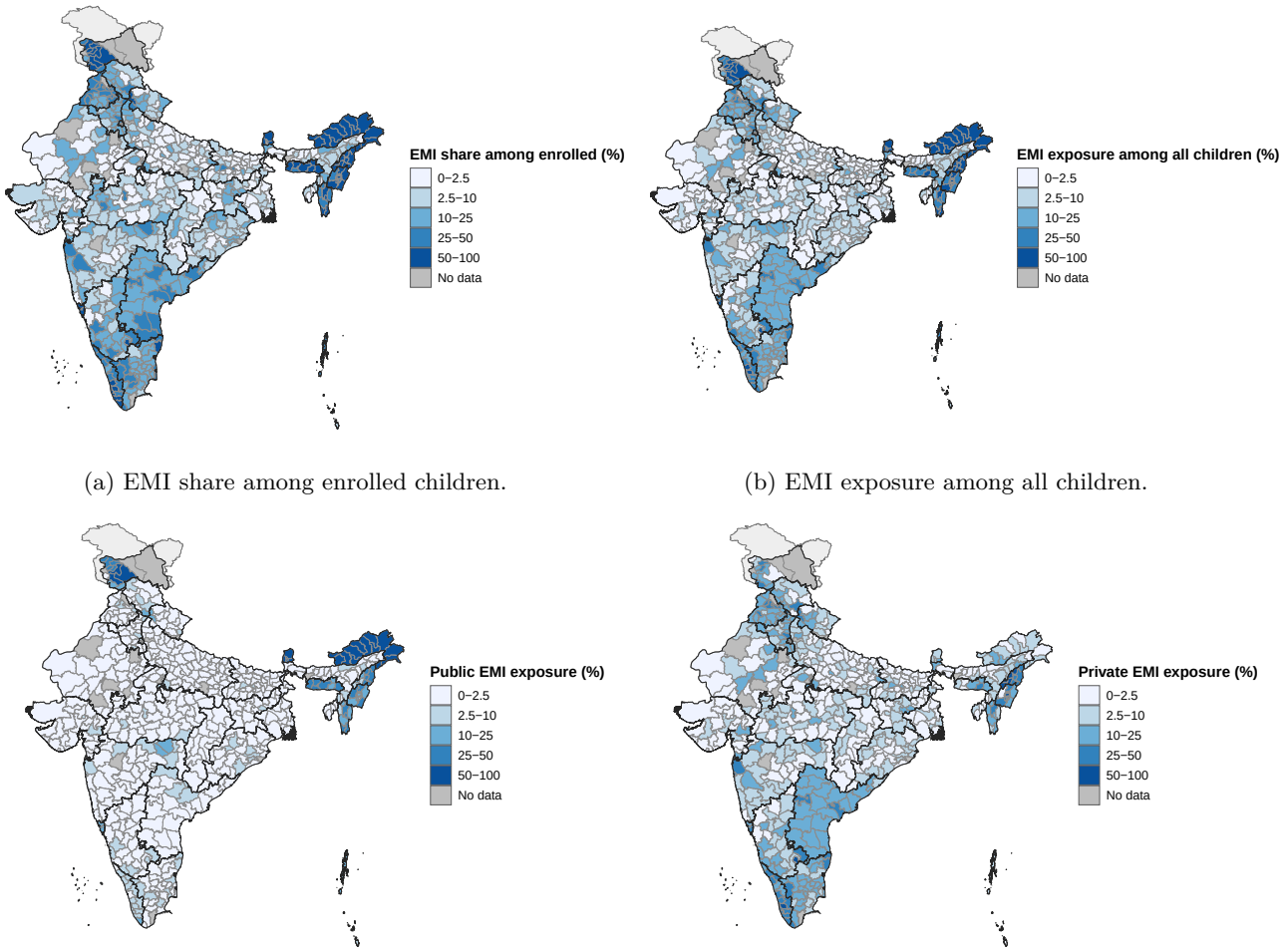
The schooling measures capture different components of realized access. Enrollment records whether an age-eligible child attends school. EMI among enrolled children records medium of instruction among children already in school, and EMI exposure among all children incorporates both enrollment and reported medium. I also separate public- and private-school EMI exposure and private-school enrollment. School management and medium of instruction identify different schooling choices: households choosing private schools often cite perceived learning quality or school environment more often than English medium itself (Joshi and Kumar 2017). I therefore keep enrollment, EMI among enrolled children, EMI exposure among all children, public EMI, private EMI, and private enrollment separate.

Reported medium of instruction is distinct from English proficiency and instructional quality. In the rural sample studied by Lahoti and Mukhopadhyay (2019), 57% of children described as attending English-medium schools were actually taught in the dominant regional language, while another 18% used English books with explanation in the regional language. The NSS and DISE measures therefore record reported school medium; they do not measure English proficiency or classroom language use.

DISE reports district enrollment by medium and management from schools. These enrollment shares reflect both

available provision and family demand. I use DISE to corroborate the geographic pattern measured in the NSS because the two datasets record the same schooling margin from different respondents (Mehta 2022; District Information System for Education 2017).

Figure 3 compares the four EMI measures directly. EMI among enrolled children isolates the medium-of-instruction margin among children already in school, while EMI exposure among all children also incorporates enrollment. Splitting exposure among all children by school management shows that public-school EMI is concentrated especially in northern and northeastern districts, whereas private-school EMI is more geographically dispersed. The four panels use the same percentage bins so that differences in both levels and geography can be compared directly.



(a) EMI share among enrolled children.

(b) EMI exposure among all children.

(c) Public-school EMI exposure among all children.

(d) Private-school EMI exposure among all children.

Figure 3: District distribution of English-medium schooling measures on 2001 Census boundaries. All panels use a common percentage scale. Source: NSS 64th Round, 2007–08 (National Sample Survey Office 2011). Tengshe (2014) and Natural Earth (2022) provide geospatial data. District boundaries reflect 2001 geography. Politically disputed regions are identified using Natural Earth, and those without data—Aksai Chin, Azad Kashmir, Gilgit-Baltistan, the Trans-Karakoram Tract, and Siachen Glacier—are colored light gray. Their inclusion does not imply a position on territorial disputes.

4.2 EMI Access as a Function of Institutional Factors

The schooling measures are organized very differently across states. Table 3 presents the same three geographic specifications for each schooling outcome: an unadjusted association, a specification with region fixed effects and predetermined 2001 Census controls, and a specification with state fixed effects and the same controls. The table reports standardized coefficients so that attenuation can be compared across outcomes, with state-clustered standard errors beneath each estimate. Nationally, linguistic distance is positively associated with every English-

medium schooling margin in the table. The association attenuates sharply within states: the standardized coefficient with state fixed effects is about 0.12 for enrollment, 0.03 for EMI among enrolled children, 0.06 for EMI exposure among all children, 0.05 for public EMI, 0.09 for private EMI, and 0.05 for DISE English-medium enrollment.

The state membership R^2 rows report the share of district variation associated with state indicators. State indicators explain roughly 84% of linguistic-distance variation. Across the schooling measures, they explain about 83–84% of EMI exposure and EMI among enrolled children, about 83% of DISE English-medium enrollment, and nearly 90% of public EMI, compared with about 57% of private EMI and 36% of enrollment. The especially strong state concentration of public EMI accords with the historical organization of Indian state boundaries and schooling systems around language. Jain (2017) shows that aligning official instructional language with local language after linguistic state reorganization substantially improved educational attainment.

Table 3: Linguistic Distance and English-Medium Schooling across Geographic Specifications

	(1)	(2)	(3)
Enrollment			
Linguistic distance from Hindi	0.249***	0.177**	0.120
	(0.082)	(0.071)	(0.079)
Region fixed effects	No	Yes	No
State fixed effects	No	No	Yes
Predetermined controls	No	Yes	Yes
Observations	573	573	573
State membership R^2	0.359	0.359	0.359
English-medium share among enrolled			
Linguistic distance from Hindi	0.364***	0.175	0.033
	(0.113)	(0.132)	(0.069)
Region fixed effects	No	Yes	No
State fixed effects	No	No	Yes
Predetermined controls	No	Yes	Yes
Observations	573	573	573
State membership R^2	0.840	0.840	0.840
English-medium exposure among all children			
Linguistic distance from Hindi	0.376***	0.190	0.063
	(0.112)	(0.134)	(0.082)
Region fixed effects	No	Yes	No
State fixed effects	No	No	Yes
Predetermined controls	No	Yes	Yes
Observations	573	573	573
State membership R^2	0.829	0.829	0.829
Public-school English-medium exposure			
Linguistic distance from Hindi	0.331**	0.176	0.055
	(0.130)	(0.126)	(0.084)
Region fixed effects	No	Yes	No
State fixed effects	No	No	Yes

Table 3: Linguistic Distance and English-Medium Schooling across Geographic Specifications *(continued)*

	(1)	(2)	(3)
Predetermined controls	No	Yes	Yes
Observations	573	573	573
State membership R^2	0.895	0.895	0.895
Private-school English-medium exposure			
Linguistic distance from Hindi	0.450***	0.190*	0.093
	(0.089)	(0.085)	(0.059)
Region fixed effects	No	Yes	No
State fixed effects	No	No	Yes
Predetermined controls	No	Yes	Yes
Observations	568	568	568
State membership R^2	0.570	0.570	0.570
DISE English-medium enrollment share			
Linguistic distance from Hindi	0.403***	0.142	0.045
	(0.112)	(0.109)	(0.057)
Region fixed effects	No	Yes	No
State fixed effects	No	No	Yes
Predetermined controls	No	Yes	Yes
Observations	520	520	520
State membership R^2	0.841	0.841	0.841

Each outcome block reports standardized coefficients from district regressions on speaker-weighted linguistic distance from Hindi. State-clustered standard errors are in parentheses. Columns (1)-(3) are unadjusted, region fixed effects plus baseline 2001 Census covariates, and state fixed effects plus the same covariates, respectively. Each outcome uses one fixed complete-case sample across its three specifications. State membership R^2 comes from a regression of each outcome on state indicators alone, using the districts available for that outcome. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Sources: NSS 64th Round, 2007–08; DISE District Report Cards, 2007–08; Census of India 2001.

DISE corroborates the NSS geographic pattern. DISE English-medium enrollment and NSS EMI among enrolled children have a raw district correlation of 0.896, while DISE and NSS EMI exposure among all children correlate at 0.882. After subtracting each state's mean from both measures, the correlations are 0.607 and 0.562. These within-state correlations remain substantial even after removing the broad state component that organizes both schooling measures. The corresponding district-level NSS–DISE comparison is shown in Figure 7.

Private schooling has a distinct household cost. Even schools described as low-fee can absorb a substantial share of poor households' income (Endow 2019), and private enrollment is strongly selected on household characteristics (Muralidharan and Kremer 2008; Raju 2023). Raw public-private achievement differences therefore do not identify differences in school effectiveness. In a randomized voucher experiment in Andhra Pradesh, Muralidharan and Sundararaman (2015) find no statistically significant difference between lottery winners and losers in Telugu, mathematics, English, or science and social studies after two and four years, despite large cross-sectional differences between public- and private-school students.

School management identifies whether enrollment is in the public or private sector. I analyze the DISE measures

of pupil-teacher ratios, single-teacher schools, sanitation, private-sector composition, and EMI separately.

4.3 EMI Access as a Function of Social Factors

Access differs sharply across social groups even within the same districts. Figure 4 compares Scheduled Tribe (ST), Scheduled Caste (SC), and Other Backward Class (OBC) children with children in the residual Other category within common districts. The social-group differences are present for enrollment but substantially larger for English-medium and private schooling. Relative to Other children, mean district enrollment is lower by about 13.6 percentage points for ST children, 9.8 points for SC children, and 6.4 points for OBC children. EMI among enrolled children is lower by about 13.3, 17.2, and 13.2 points respectively. Private enrollment is lower by about 18.1 points for ST children, 23.5 for SC children, and 14.5 for OBC children. Household resources and social institutions can shape these schooling choices (Endow 2019; Raju 2023). In Mumbai, Munshi and Rosenzweig (2006) show that caste networks continued to direct lower-caste boys toward local-language schools associated with traditional occupations as returns to nontraditional white-collar work rose, while lower-caste girls shifted more rapidly toward English schools.

The rural/urban comparison makes the social stratification more pronounced. For SC children, for example, the difference in EMI among enrolled children is about 8.4 percentage points in rural districts and 24.6 points in urban districts; the corresponding difference in private EMI exposure is about 7.5 and 21.4 points. OBC and ST children show the same qualitative pattern. Urban schooling markets offer a wider range of formal options while household resources and social position continue to shape which options are reachable. This pattern describes sorting across households within urban districts.

The corresponding decompositions for boys and girls are similar. Government programs operating before the NSS round explicitly targeted girls' education (Press Information Bureau 2008), so I also examine the sex split directly. In this national cross-section, social group and rural/urban setting distinguish access more strongly than sex.

English-medium and private schools can be present in a district while remaining unequally accessible across households. The next section asks whether these observed EMI choices are associated with later economic outcomes that might help explain household demand for them.

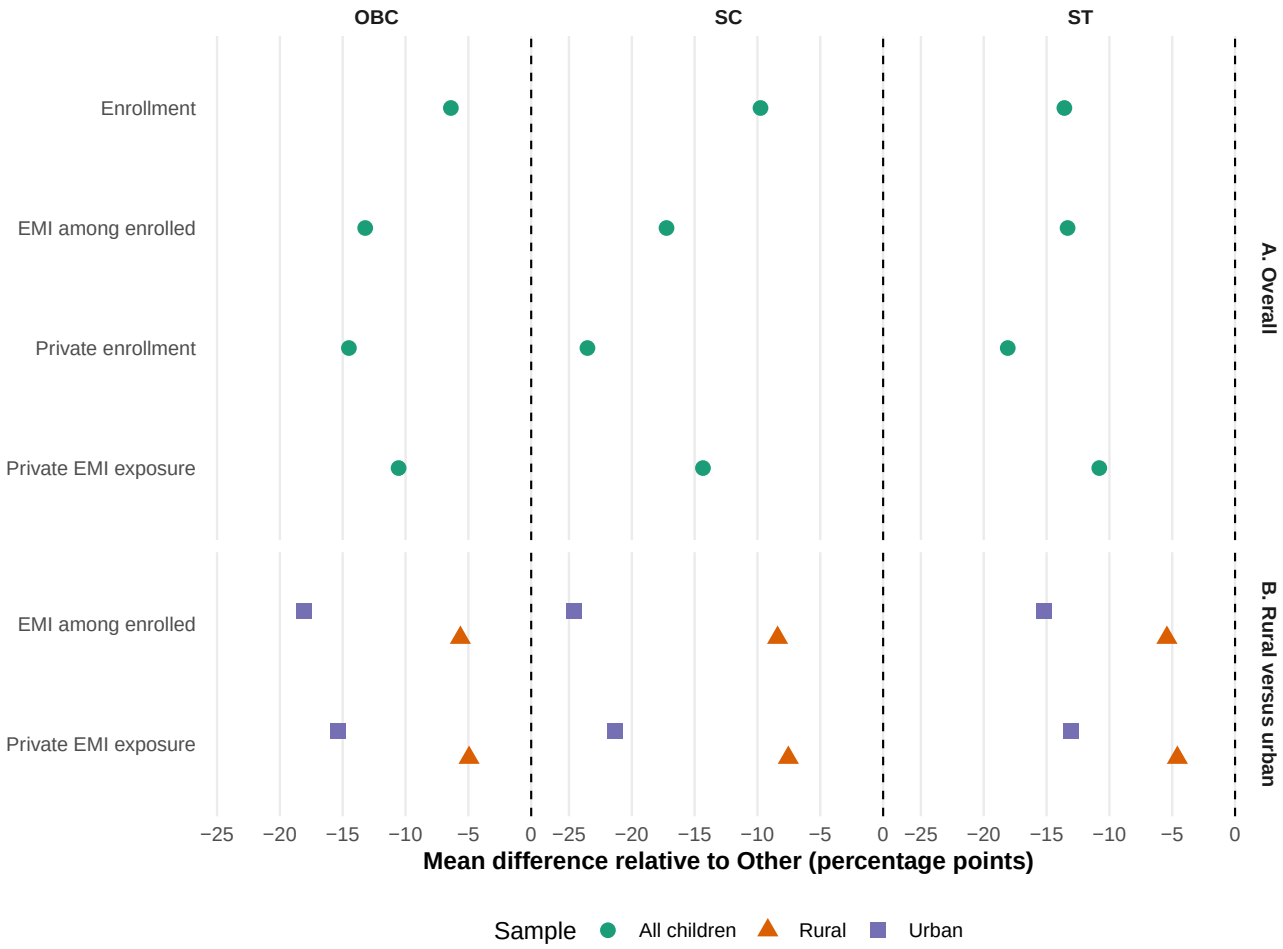


Figure 4: Social-group differences in access to English-medium and private schooling. Panel A reports mean within-district percentage-point differences for OBC, SC, and ST children relative to the Other category in enrollment, EMI among enrolled children, private enrollment, and private EMI exposure among all children. Panel B separates rural and urban differences for EMI among enrolled children and private EMI exposure. Source: NSS 64th Round, 2007–08 (National Sample Survey Office 2011).

5 Local Economic Outcomes Associated with English-Medium Instruction

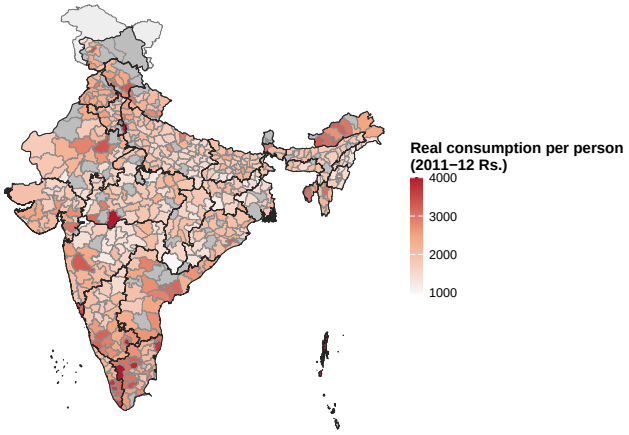
5.1 Later Consumption and English-Medium Instruction

English-language skills are a form of human capital (Chiswick and Miller 1995; Azam et al. 2013), and evidence from India gives households an economic reason to value them. Using nationally representative 2005 household data, Azam et al. (2013) find English fluency associated with a 34% hourly wage premium for men and a 22% premium for women; workers with a little English receive smaller positive premia, and the returns are greater for more educated workers. Refeque and Azad (2022) later documents substantial English-skill earnings associations through 2011–12. The strongest evidence for a causal wage effect comes from Chakraborty and Bakshi (2016), who exploit West Bengal's withdrawal of English from primary schools and estimate substantially lower later wages for cohorts more exposed to the policy.

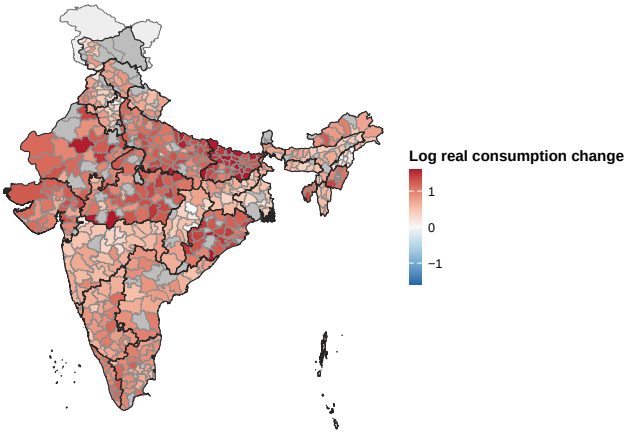
Households making schooling decisions in 2007–08 could observe contemporary labor-market rewards to English, including the wage differences documented in this literature. I use later district consumption to describe the economic outcomes eventually associated with the schooling choices observed in 2007–08.

Later district consumption varies substantially across India. Figure 5 shows real consumption per person in 2022–23 and the long change from 2004–05 to 2022–23 among districts satisfying the preferred support criteria for both rounds. The figure shows the raw geographic distribution. The regressions below condition on state fixed effects and baseline 2001 Census covariates.

Table 4 compares five schooling margins on common district support. Each entry comes from a separate district regression with state fixed effects and baseline 2001 Census covariates. The table places the coefficients on the same scale for comparison. In the 2022–23 ANCOVA specification, a 10 percentage-point increase in enrollment is associated with 0.7% higher real mean consumption (SE 1.70). The corresponding estimates are 3.9% for EMI among enrolled children, 5.1% for EMI exposure among all children, 4.1% for public EMI, and 5.3% for private EMI. The EMI among enrolled children, EMI exposure among all children, and private EMI estimates survive the predeclared family correction. In the 2004–05 to 2022–23 long-run change specification, EMI exposure among all children is associated with 3.9% higher real mean consumption and private EMI with 5.4%; both survive family correction. The analogous 2023–24 estimates are smaller, and none survives the predeclared family correction.



(a) Real consumption per person, 2022–23.



(b) Log change in real consumption per person, 2004–05 to 2022–23.

Figure 5: Real district consumption in 2022–23 and its long-run change from 2004–05 to 2022–23. Both panels use districts observed in the two comparison rounds. Real consumption is expressed in 2011–12 rupees. Sources: NSS 61st Round, 2004–05, and HCES 2022–23 (National Sample Survey Office 2007, 2024b). Tengshe (2014) and Natural Earth (2022) provide geospatial data. District boundaries reflect 2001 geography. Politically disputed regions are identified using Natural Earth, and those without data—Aksai Chin, Azad Kashmir, Gilgit-Baltistan, the Trans-Karakoram Tract, and Siachen Glacier—are colored light gray. Their inclusion does not imply a position on territorial disputes.

Table 4: English-Medium Schooling, Later Consumption, and Predetermined Complements

	(1)	(2)	(3)	(4)
Enrollment	0.70	1.27	-0.02	0.72
	(1.73)	(1.44)	(1.09)	(1.47)
EMI among enrolled	3.88*	2.79	2.31	0.82
	(1.15)	(1.13)	(1.14)	(1.22)
EMI exposure among all children	5.06**	3.88*	2.62	1.06
	(1.40)	(1.22)	(1.44)	(1.61)
Public EMI	4.07	1.13	2.04	-0.94
	(1.46)	(2.26)	(1.15)	(2.56)
Private EMI	5.31*	5.44***	2.94	2.53
	(1.71)	(1.30)	(1.99)	(1.82)
EMI exposure among all children × Baseline human capital		1.50*		
		(0.47)		
EMI exposure among all children × Urbanization		1.83*		
		(0.63)		
EMI exposure among all children × ST concentration		0.04		
		(0.55)		
Private EMI × Baseline human capital		1.47		
		(0.82)		
Private EMI × Urbanization		2.00		
		(1.02)		
Private EMI × ST concentration		0.40		
		(0.83)		
Endpoint	HCES 2022-23	HCES 2022-23	HCES 2023-24	HCES 2023-24
Estimand	ANCOVA	2004-05 to 2022-23 change	ANCOVA	2004-05 to 2023-24 change
State fixed effects	Yes	Yes	Yes	Yes
Predetermined controls	Yes	Yes	Yes	Yes
Observations	524	524	522	522

Each schooling-margin row comes from a separate district regression and reports the percent difference in real mean MPCE associated with a 10 percentage-point increase in that schooling margin. Rows sharing a column are displayed together for comparison. Columns include state fixed effects, baseline 2001 Census covariates, and a common district sample within each outcome specification. State-clustered standard errors are in parentheses; stars use Holm-adjusted p-values within the prespecified schooling-welfare family. The six interaction rows apply only to the 2004-05 to 2022-23 long-run change specification. They report how the coefficient on EMI exposure among all children or private EMI changes with a one-standard-deviation increase in baseline human capital, urbanization, or ST concentration. Each interaction comes from a separate regression; stars use Holm-adjusted p-values within the prespecified complement family. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The coefficients are descriptive district associations; the interaction rows describe slope heterogeneity across baseline district characteristics. Sources: NSS 64th Round, 2007–08; Census of India 2001; NSS 61st Round, 2004–05; HCES 2022–23; HCES 2023–24.

These are conditional district associations, while individual wage studies estimate returns to English skills for workers. The strong social selection into English-medium and private schooling makes that distinction important. EMI exposure is much more strongly associated with later consumption than enrollment alone, consistent with Azam et al. (2013)'s evidence that English skills and education are complementary in the labor market.

5.2 Predetermined Complements to English-Medium Instruction

The same Table 4 asks whether the 2004–05 to 2022–23 association varies with three characteristics measured before the schooling round: baseline human capital, urbanization, and ST concentration. The interaction rows standardize each characteristic so that they report how much the schooling association differs with a one-standard-deviation increase. For EMI exposure among all children, the schooling–consumption association is about 1.5 percentage points larger for a one-standard-deviation increase in baseline human capital and about 1.8 points larger for the same increase in urbanization. Both interactions survive the predeclared family correction. The ST-concentration interaction is near zero. The private EMI interaction estimates are also positive: 1.47 points for baseline human capital and 2.00 for urbanization.

For EMI exposure among all children, the association with consumption is larger in districts with higher baseline human capital and urbanization. The predeclared IT-employment interaction is near zero for both EMI exposure and linguistic distance.

The EMI margins most strongly associated with later district consumption are also the schooling margins that are sharply stratified by social group. The association between EMI exposure among all children and consumption growth is larger where baseline human capital and urbanization are greater. This pattern links unequal access to EMI with unequal baseline conditions associated with a stronger later consumption relationship.

5.3 Economic Outcomes as a Function of Linguistic Conditions

The final part of the local-outcomes analysis examines whether inherited linguistic conditions are associated with later economic change.

The consumption association appears over the long horizon. Figure 6 follows the coefficient on linguistic distance across the later-consumption endpoints. The 2009–10 and 2011–12 estimates are small relative to their uncertainty. The 2004–05 to 2022–23 long-run change estimate is positive and precisely estimated, and the 2023–24 estimate remains positive with greater uncertainty. The local development outcomes below show which district characteristics change alongside this later consumption association.

Table 5 reports the predeclared local development outcomes spanning human capital, migration, household financial access, sectoral composition, and employment. The point estimates are positive for household literacy depth, graduate access, skilled recent-work in-migration, banking access, television ownership, and the services employment share, and negative for the manufacturing employment share. These directions are descriptive: none of the displayed associations survives the prespecified Holm correction within its outcome family. The migration outcome measures the skill composition of recent work in-migrants at their destination, while the interstate share among migrants is close to zero in the preferred specification.

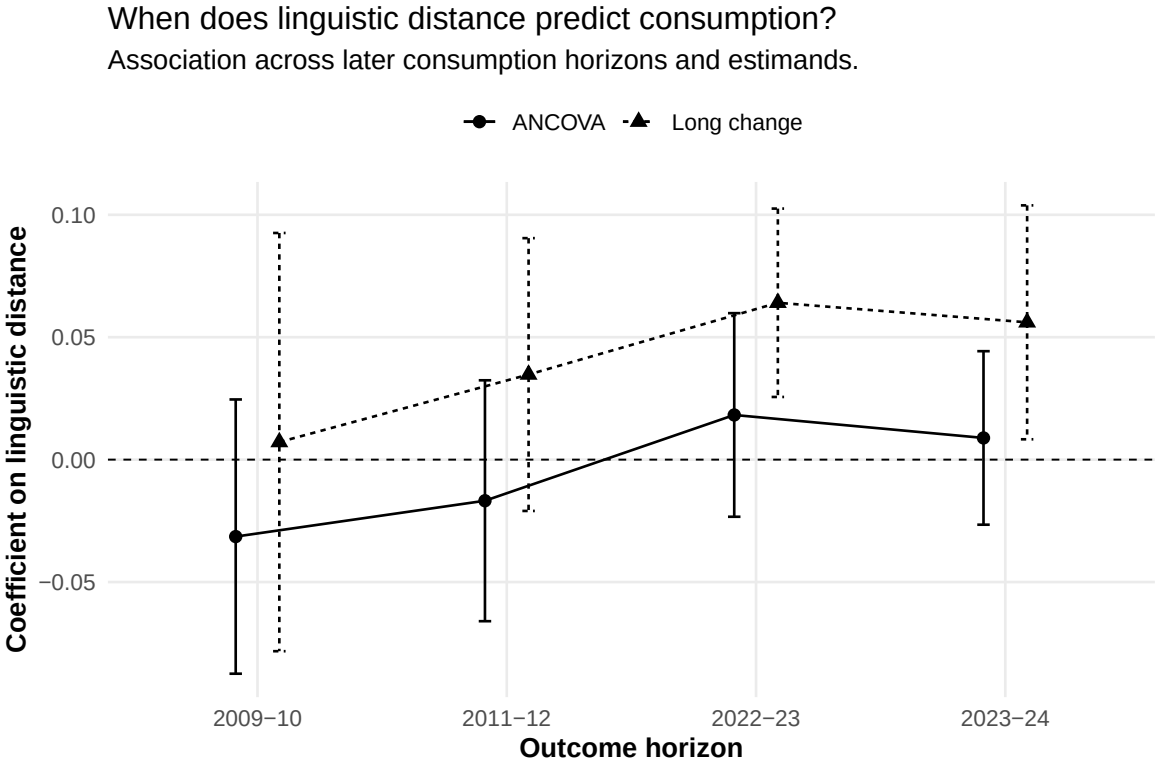


Figure 6: Association between linguistic distance and real mean household consumption across survey horizons. Points show ANCOVA and long-run change coefficients with 95% state-clustered Wald confidence intervals; all specifications include state fixed effects and baseline 2001 Census covariates. Sources: Census of India 2001; NSS 61st, 66th, and 68th Rounds; HCES 2022–23 and 2023–24 (Office of the Registrar General and Census Commissioner of India 2001; National Sample Survey Office 2007, 2012b, 2014, 2024b, 2025a).

Table 5: Linguistic Conditions and Local Development Outcomes

	(1)	(2)	(3)	(4)
Panel A: Household capacity and assets				
Linguistic distance	0.0092	0.0041	0.0475	0.0141
	(0.0040)	(0.0017)	(0.0148)	(0.0060)
Dependent variable	2+ literates	Graduate access	Banking access	Television ownership
Period	2001-11	2001-11	2001-11	2001-11
Fixed effects	State	State	State	State
Predetermined controls	Yes	Yes	Yes	Yes
Observations	355	355	355	355
Panel B: Migration and sectoral composition				
Linguistic distance	0.0129	-0.0023	0.0206	-0.0179
	(0.0049)	(0.0199)	(0.0087)	(0.0083)
Dependent variable	Skilled recent-work migrant share	Interstate migrant share	Services employment share	Manufacturing employment share
Period	2011	2011	2005-13	2005-13
Fixed effects	State	State	Region	Region
Predetermined controls	Yes	Yes	Yes	Yes
Observations	355	355	354	354
Panel C: Employment scale and labor outcomes				
Linguistic distance	-0.0226	-0.0074	-0.0085	
	(0.0255)	(0.0064)	(0.0063)	
Dependent variable	Log nonfarm employment	Labor-force participation	Employment rate	
Period	2005-13	2017-18	2017-18	
Fixed effects	Region	State	State	
Predetermined controls	Yes	Yes	Yes	
Observations	354	467	467	

Each column is a separate district regression. The coefficient reports the association of one Shastry linguistic-distance degree with the listed outcome in its native unit; state-clustered standard errors are in parentheses and stars use Holm-adjusted p-values within the corresponding outcome family. Census household, migration, finance, asset, and labor regressions use state fixed effects plus baseline 2001 Census covariates. Economic Census regressions use region fixed effects plus the same covariates, as reported in the table. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The 2001-11 Census and 2005-13 Economic Census windows overlap the 2007 schooling measurement, so I interpret them as contemporaneous dimensions of district development. PLFS 2017-18 supplies the displayed labor outcomes. Outcomes remain in their native units. Sources: Census of India 2001; Census of India 2011; Fifth Economic Census 2005; Sixth Economic Census 2013–14; NSS 66th Round; PLFS 2017–18.

The local development point estimates are concentrated in human capital, skilled in-migration, financial access, and sectoral composition, but the family-adjusted inference is weak. The alternative welfare measures are less consistent than real mean MPCE. In the Economic Census, the services-share estimate is positive and the manufacturing-share estimate negative, while total nonfarm employment changes little. Changes in labor-force participation and the employment rate are also small relative to their uncertainty. I therefore use these outcomes to describe the composition of the local-development pattern rather than as individually precise findings.

The signs of the sectoral point estimates resemble the service- and skill-intensive pattern documented in India's broader development literature. Kochhar et al. (2006) emphasize the country's service- and skill-intensive pattern

relative to the manufacturing-led path followed by several other fast-growing Asian economies. More recent work stresses how tradeable services can become increasingly important to development as technology changes the tradability of service work (Baldwin and Forslid 2023). In this district analysis, however, the family-adjusted evidence does not isolate any one local-development outcome precisely enough to support a separate mechanism claim.

The Census household and housing outcomes cover 2001–11, while the Economic Census changes cover 2005–13; both windows overlap the 2007–08 schooling measurement. I therefore treat these outcomes as contemporaneous district characteristics around the schooling measurement. The schooling–consumption association is stronger in districts with greater baseline human capital and urbanization. The local-development estimates provide descriptive context on human capital, migration composition, financial access, and sectoral composition, with substantial uncertainty after family correction.

6 Conclusion

English skills are often presented as a way for households to loosen the connection between children's prospects and the circumstances into which they are born (Itani et al. 2015; Bedi 2019). The premise is economically plausible: English proficiency carries substantial labor-market rewards in India (Azam et al. 2013; Refeque and Azad 2022), and English schooling has been associated with access to occupations outside traditional caste-linked networks (Munshi and Rosenzweig 2006). I began from a narrower question. If English-medium schooling is to expand access to valuable English skills, what conditions surround a household's access to it, and are those conditions themselves related to inherited and social inequality?

The evidence points to inequality in each of the three conditions studied in this paper. The linguistic conditions I call inherited are highly persistent across the 1991 and 2001 reconstructions. Their association with reported English and Hindi differs substantially across settings. State schooling systems account for most of the geographic variation in EMI exposure and provision: state membership explains about 84% of the variation in EMI among enrolled children and about 90% of government-school EMI exposure. Within the same districts, realized access to English-medium and private schooling differs sharply across social groups.

The later consumption results show a corresponding economic pattern. In the 2022–23 ANCOVA comparison, a 10 percentage-point increase in EMI exposure among all children is associated with 5.1% higher real mean MPCE, compared with 0.7% for enrollment. The estimates for EMI among enrolled children, EMI exposure among all children, and private EMI survive the predeclared family correction. In the long-run change specification, the estimates for EMI exposure among all children and private EMI also survive that correction. The schooling–consumption association is larger in districts with greater baseline human capital and urbanization. Greater

linguistic distance is also associated with gains in household literacy and household graduate access, a more skilled composition of recent work in-migration, greater banking access, and a shift in employment toward services. India's service- and skill-intensive development path provides context for the sectoral pattern (Kochhar et al. 2006; Baldwin and Forslid 2023).

The identification analysis reaches a clear limit. Under the preferred specification with state fixed effects, linguistic distance has a partial R^2 of about 0.004 for EMI exposure among all children, and the IV variants fail the effective-F strength criterion. Alternative linguistic-distance measures also remain weak after state adjustment. Accordingly, linguistic distance remains a descriptive measure of inherited linguistic conditions in the main analysis. This result also reinforces the substantive finding that state schooling systems account for most of the observed geographic variation in EMI.

Households begin investment in English skills from unequal positions. Inherited linguistic conditions differ across districts, realized access to EMI differs sharply by social group within the same districts, and the later consumption relationship is larger where baseline human capital and urbanization are higher. A high-return educational opportunity can therefore coexist with substantial inequality in access and in the local conditions associated with its later economic returns.

7 Technical Note for Replication

All reported tables and figures are generated through the project's `{targets}` build. Exact R package versions are recorded in `renv.lock`. Required raw files are declared in `data/metadata/file_manifest.csv` with expected paths and sizes; the manifest also records SHA-256 identities for verified inputs. `DATA_AVAILABILITY.md` records availability and redistribution constraints for each required dataset. The appendices document data construction (Section A), migration (Section B), selection (Section C), and weak identification (Section D). The repository also provides machine-readable results for the larger validation and local development analyses. `REPLICATION.md` gives the reproduction procedure and system requirements. The full build checks package synchronization and parses the active R and Quarto files before running the test suite. It then renders the manuscript, verifies cross-references and required outputs, and checks reproduction from the tracked processed district inputs.

A Major Data Contributions

The paper relies on original longitudinal datasets that I construct to make records from different years and collections comparable. I place district observations on a common 2001 Census geography, reconstruct historical DISE medium-of-instruction measures, build district linguistic measures from Census mother-tongue populations and historical language records, and construct comparable real consumption estimates from nine survey designs spanning 2000–01 through 2023–24. I also construct the Census and Economic Census panels used for the local development outcomes. This appendix describes each construction and its validation.

A.1 District Lineage Harmonization

District names cannot serve as longitudinal identifiers in India. Between the 1961 and 2001 Censuses the number of districts rose from 339 to 593 through renamings, partitions, mergers, and boundary changes (Kumar and Somanathan 2016). The 2011 Census contained 640 districts after another 47 were created during the intervening decade (Office of the Registrar General and Census Commissioner, India 2012). I therefore construct a reviewed geographic concordance that records how districts observed in later surveys relate to the 593 2001 Census districts used as the reference geography.

Reference geography and evidence. The 593 2001 Census districts provide the common geographic basis for the paper. Historical boundary evidence extends back to 1961, and later survey links extend through the 2023–24 HCES round. Kumar and Somanathan's district histories provide the benchmark for pre-2001 partitions and population transfers (Kumar and Somanathan 2016). Census records and the Ministry of Panchayati Raj's Local Government Directory establish later district identities, creations, renamings, and code continuity (Ministry of Panchayati Raj, Government of India 2026). The India State and District Evolution Database, the India District Changes Tracker, and published survey concordances provide additional evidence on district histories and survey-to-Census links (Jolad et al. 2026; Jaacks et al. 2020; Deshpande et al. 2024). SHRUG locality keys measure population overlap between 2001 and 2011 places when later districts cross 2001 boundaries (Asher et al. 2021).

Establishing district identity. Standardized names and spellings generate candidate matches. Accepted identities require documentary or geographic evidence. Each survey district is tied to a Census or LGD unit using its code, an exact or documented renamed identity, or a reviewed record. The review table records the decision and supporting evidence for ambiguous cases. Across the 2007–08 and 2017–18 NSS schooling surveys, I review 1,259 district records and use political documentation and geographical evidence to establish the contemporaneous administrative identity of 1,254: 588 from 2007–08 and 666 from 2017–18. District reorganizations are recorded as dated predecessor–successor relationships so that splits, mergers, renamings, and unchanged districts remain distinguishable.

Assigning later districts to the 2001 Census. I trace each accepted later unit to its 2001 Census predecessor or predecessors. Unchanged districts and documented renames receive unit weight on the corresponding 2001 district. A later district formed wholly within one 2001 district can also be assigned to that predecessor when the historical evidence establishes a single-parent relationship. Multi-parent districts remain outside the primary analysis. For the expanded sensitivity sample, SHRUG locality populations provide fractional weights across 2001 predecessors; accepted weights are nonnegative and sum to one within each later district. This follows the population-weighting principle used by Kumar and Somanathan (2016) for historical district partitions.

Primary and sensitivity samples. The *conservative* lineage variant includes direct identities, documented renames, and deterministic one-parent transitions. The *primary* variant also includes reviewed single-parent cases supported by Census or LGD code continuity or by independent district-history evidence corroborated with SHRUG locality overlap. Cases using locality overlap require at least 99 percent population coverage. The *expanded* variant additionally includes accepted fractional allocations across multiple 2001 predecessors. Across the two schooling waves, these definitions produce 427 common districts under the conservative variant, 573 under the primary variant used in the main analysis, and 587 under the expanded variant.

Aggregating district measures. Counts are summed across later units assigned to a common 2001 district. Rates, shares, and means are reconstructed from pooled numerators and denominators. Distributional measures such as the consumption Gini are reconstructed from pooled household records when the microdata permit it. These rules preserve the population represented by each observation when district boundaries change.

Geographic boundaries. The geographic files use 2001 Census district boundaries. The available 2001 boundary file omits several districts whose boundaries remained unchanged into 2011. I carry back a 2011 polygon only when the Census atlases establish exact continuity (Office of the Registrar General and Census Commissioner, India 2001, 2012). I use this rule for eleven districts. The same district identities therefore govern the statistical concordance and the geographic displays.

A district observation has the same interpretation across the three lineage variants; only geographic coverage changes. The identification conclusion is unchanged across the conservative, primary, and expanded variants.

A.2 DISE Data Reconstruction

Recovering the historical files. I reconstruct historical District Information System for Education (DISE) data from archived workbooks and published district report cards. DISE records schools directly and therefore provides an independent measure of school-system medium of instruction. The original publication site is no longer maintained, and file structures change across years. The surviving publication index maintained by Arun Mehta identifies the district report cards and raw files released by NIEPA/NUEPA and records 2005–06 as

the first year with nationwide DISE coverage under a common data-capture format (Mehta 2022). Using that index, I recovered an Internet Archive snapshot of the former DISE district-report-card site together with the corresponding workbooks and report cards (District Information System for Education 2017). The local archive spans 2002–03 through 2015–16. The 2005–06 through 2007–08 files provide the baseline measures used in the paper, and the later years support longitudinal checks.

Reading year-specific files. Each academic year has a documented specification for the archived file and any contained ZIP member. It identifies the relevant enrollment, school, and teacher sheets, the associated report-card volumes, and known file problems. These specifications are necessary because workbook layouts and file extensions do not identify a stable location for each measure across vintages. The baseline construction begins in 2005–06, when nationwide coverage becomes comparable (Mehta 2022).

Decoding medium of instruction. The 2005–08 enrollment workbooks contain up to five numbered medium slots with grade-specific pupil counts. The slot number does not identify the language. I sum grade counts within each slot and recover the language identity from the corresponding district report card. This reconstruction yields 7,211 district-year-slot language identities, each linked to a report-card PDF and page. English enrollment is the sum of slots identified as English, and Hindi enrollment is constructed in the same way. Slots with positive enrollment but unidentified language remain unclassified. When the English slot can be identified, its pupil count remains usable; unresolved positive slots leave English enrollment missing when they prevent identification of the English count.

Reconstructing enrollment and school characteristics. I reconstruct total Classes I–VIII enrollment separately from the medium slots. The baseline sequence uses grade totals when available, followed by the workbook's direct enrollment total and then the sum of government and private enrollment. Medium-classified enrollment is retained as a consistency measure. English-medium enrollment share equals English-medium pupils divided by total elementary enrollment. I separately sum government and private management categories to construct private enrollment and private-school shares. School and teacher files provide school counts, teacher counts, single-teacher-school counts, and girls'-toilet counts; ratios are calculated after assembling the underlying counts.

School-age population denominators. Population-scaled measures use Census counts of children ages 6–13. I first place the 2011 age counts on 2001 Census geography using deterministic district links and then use the 2001 and 2011 anchors to estimate annual district population growth on a log scale. Intervening academic-year populations are obtained by log-linear interpolation. The English-medium gross enrollment ratio divides English-medium pupils by the resulting age-6–13 population. The 2005–08 baseline measures use interpolation. Years after 2011 are used only for longitudinal checks.

District harmonization. DISE pupil and school counts use a stricter geographic rule than the general lineage

system. A DISE district is assigned to 2001 Census geography only through an exact identity or a reviewed one-to-one link with weight one. Several later districts can be pooled into one 2001 district when each link is deterministic. Pupil, school, and teacher counts are summed before recomputing shares and ratios.

Checks against published DISE reports. Published report cards both identify medium slots and validate selected reconstructed cells. Three 2005–06 medium-of-instruction cells for Kupwara reproduce the published report exactly. Later report-card pages provide English and Hindi pupil counts when workbook slot identities are unstable. The 2009–10 archive lacks its second report-card volume, so the reconstruction uses the archived complete *District Elementary Education Report Card: 2009–10* for affected pages (National University of Educational Planning and Administration 2012). The 2010–11 enrollment workbook has a documented header/data alignment problem; matched published district totals provide the current-year denominator in affected cases. These year-specific rules are defined before estimation.

Independent comparison with NSS. The independently reconstructed DISE and NSS measures agree strongly in district levels. For 2007–08, the correlation between DISE English-medium enrollment share and NSS English-medium instruction among enrolled children is 0.896 across 520 districts. After subtracting state means from both series, the correlation remains 0.607. The within-state correlation shows substantial agreement between two independently constructed measures after removing their common state component.

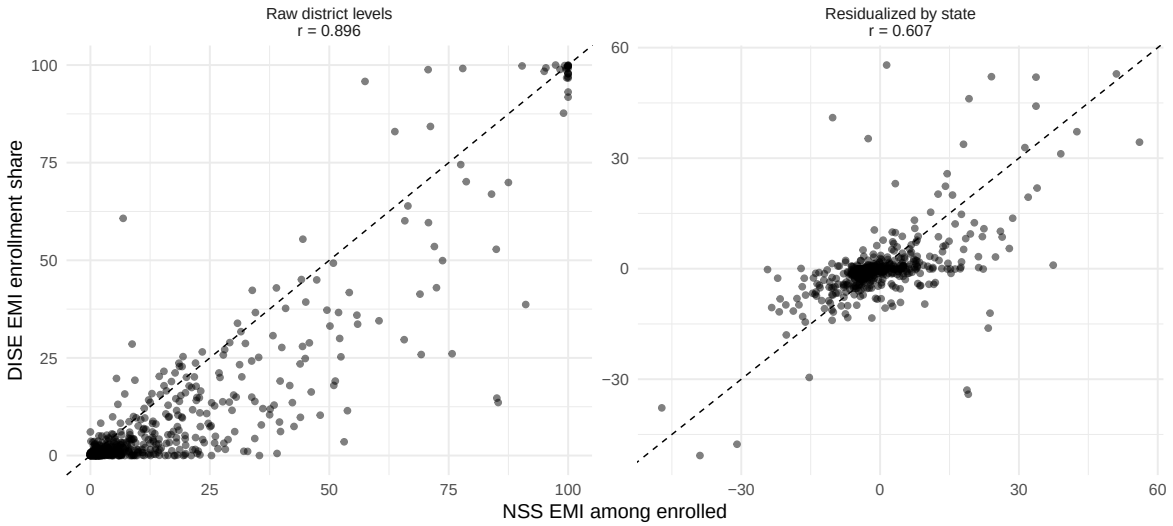


Figure 7: Agreement between NSS and DISE English-medium measures. The left panel compares district levels; the right panel subtracts each source's state mean before comparison. Sources: NSS 64th Round, 2007–08 (National Sample Survey Office 2011) and DISE District Report Cards, 2007–08 (District Information System for Education 2017).

A.3 Linguistic Measures

The linguistic measures summarize differences in the relative cost of acquiring Hindi and English as languages of wider communication, following Shastry (2012). I begin with mother-tongue-level distance from Hindi and combine those scores with each district's distribution of mother-tongue speakers.

Census speaker population. The speaker distribution comes from the mutually exclusive mother-tongue rows of Census 2001 Table C-16 (Office of the Registrar General and Census Commissioner of India 2001). Language-group subtotals are excluded from the speaker count. This treatment is important for groups such as Census Hindi, which contains distinct mother tongues including Bhojpuri and Rajasthani varieties. Distance is assigned at the mother-tongue level before district aggregation.

Shastry distance measures. The preferred language scores use Shastry's ordinal 0–5 classification (Shastry 2012; see also Shastry 2008, fig. 5). Hindi and Urdu form the zero-distance reference. Published Indo-European languages retain Shastry's reported degree. Languages outside Indo-European receive degree five after review of their family classification. Native English is handled separately and excluded from the weighted-distance mean. The primary *speaker-weighted distance from Hindi* is the Census-speaker-weighted mean among classified mother tongues with positive Shastry distance. The older *top-three-language distance* uses only the three largest classified mother tongues in each district. The primary measure uses the full classified mother-tongue distribution; I retain the top-three measure for comparison with the earlier analysis.

Language classification review. Some Census mother tongues have no unambiguous published Shastry category. I apply the published Shastry concordance first. Finer Indo-European mother tongues absent from that concordance are classified using historical evidence from the *Linguistic Survey of India* (Grierson 1903–1928) and a named published Shastry anchor. Lexical evidence from Kogan (2017) and ASJP v21 (Wichmann et al. 2025) helps distinguish nearby classifications when needed. The reviewed classification table records accepted assignments and unresolved rows. Mother tongues without sufficient evidence remain unclassified in the preferred construction. Regression results play no role in these classification decisions.

Population shares and unresolved languages. Unclassified speaker mass is retained explicitly. The preferred weighted mean uses classified positive-distance speakers in its numerator and denominator, with the classified-speaker share reported separately. Population-share measures keep unresolved and native-English speakers in the district population denominator. The *share speaking languages distant from Hindi* is the percentage of all mother-tongue speakers whose classified language is at least three Shastry degrees from Hindi. Five additional shares record district speaker mass at each positive distance degree, with distance zero serving as the omitted category when the shares enter jointly. Low- and high-degree assignments for reviewed ambiguous languages provide bounding sensitivity checks while the preferred assignments remain fixed.

Historical 1991 measure. I reconstruct a separate 1991 district-language series from the *Language Atlas of India 1991*, which reports the geographic distribution of 114 classified languages (Office of the Registrar General and Census Commissioner, India 2004). The Atlas is distributed as a PDF, so I extract district-by-language counts from the text and page coordinates embedded in its district sheets. Ambiguous cells are recorded for review and remain unresolved until supported by the page content. Districts are identified using the 1991 Census state and district codes (Registrar General and Census Commissioner, India 1991). Extracted district populations are checked against the 1991 Primary Census Abstract before language shares are calculated. Atlas languages are then linked to the same Shastry basis, and accepted district-language counts are checked against Atlas population totals.

Unresolved language mass enters explicit bounds on the weighted mean. The preferred historical series requires at least 99 percent accepted speaker coverage and a distance-bound width no greater than 0.5 degree. Eligible 1991 districts are then placed on 2001 Census geography using Section A.1. The resulting series supports the 1991–2001 persistence analysis and the independent reconstruction checks reported in the main text.

Genealogical and lexical alternatives. Two alternative constructions use different linguistic information. The *genealogical distance from Hindi* uses the reviewed Census-to-Glottolog 5.3 crosswalk and counts separation along Glottolog's language-family genealogy (Hammarström et al. 2026), with Shastry's published Indo-European tree retained as an external interpretation check (Shastry 2008, fig. 6). The edge count remains on its native genealogical scale. The *lexical noncognacy with Hindi* measure uses the Dyen–Kruskal–Black Indo-European cognate data (Dyen et al. 1992; "Comparative Indo-European Database" 1997). At the language level it equals one hundred minus the percentage of cognate items shared with Hindi. Following Shastry's convention, reviewed non-Indo-European languages outside the Dyen comparison receive five percent shared words, giving a noncognacy value of 95. Unresolved Indo-European mother tongues remain unresolved. All three distance constructions are defined independently of their regression results.

A.4 Price Series and Consumption Harmonization

I construct comparable district consumption estimates from nine survey designs spanning NSS 56 in 2000–01 through HCES 2023–24, including both NSS 66 questionnaire designs. The main longitudinal analysis uses NSS 61 as the baseline, NSS 66 Type 2 and NSS 68 Type 2 as intermediate endpoints, and the two HCES rounds as long-run endpoints. The other survey designs provide pretrend, treatment-period, and questionnaire-design checks. Survey questionnaires, recall periods, district codes, and price environments differ across these rounds (National Sample Survey Organisation 2002, 2003; National Sample Survey Office 2007, 2012a, 2012b, 2014, 2024b, 2025a; National Statistical Office 2011). I reconstruct household monthly per-capita consumption expenditure

(MPCE) within each design, validate nominal MPCE against published survey totals, and then harmonize prices and district geography. The paper uses real consumption per person across rounds.

NSS consumption definitions. For each NSS Schedule 1.0 design, I retain its published consumption concept. The metadata record the household MPCE field, scale, survey-design variables used for weighting, and whether the published benchmark uses the mixed reference period (MRP) or modified mixed reference period (MMRP). Under MRP, less frequently purchased categories use a 365-day recall and most other spending uses 30 days; MMRP additionally uses a 7-day recall for selected food and related items (National Sample Survey Office 2025b). Some releases report MPCE directly, while others require reconstruction from split household files. All 18 reconstructed rural and urban benchmark means across the nine surveys differ from their published counterparts by less than 0.50. A survey round enters the district series only after passing its national rural and urban MPCE benchmark.

HCES three-visit reconstruction. The modern HCES releases divide consumption across three questionnaires administered in separate visits: the Food Questionnaire (FDQ), Consumables and Services Questionnaire (CSQ), and Durable Goods Questionnaire (DGQ) (National Sample Survey Office 2024b, 2025b). Items use 7-, 30-, or 365-day reference periods. I convert each reported expenditure amount to a monthly equivalent and sum the three questionnaire components by household. Imputed house and garage rent is excluded in accordance with the published MPCE concept used for benchmarking. The 2023–24 release identifies visit order, so I use the third-visit multiplier prescribed by the official estimation procedure (National Sample Survey Office 2025b). The 2022–23 household is retained when its three questionnaire records carry the same multiplier, which supplies the common household weight. A complete HCES consumption observation requires all three questionnaire components.

Temporal price adjustment. The *temporal price relative* measures price change across survey dates within a state and rural or urban sector. Before 2013, rural prices use the Labour Bureau's Consumer Price Index for Rural Labourers (CPI-RL), whose 1986–87-base series covers 20 states and all-India (Labour Bureau, Government of India 2026b). Urban prices use the Labour Bureau's 2001-base Consumer Price Index for Industrial Workers (CPI-IW), aggregated from centre series to state level (Labour Bureau, Government of India 2026a). From 2013 onward, I use state rural and urban CPI series with base 2012=100 from the national CPI system (Reserve Bank of India 2026). Common state-sector months in 2013–14 link the earlier series to the 2012-base CPI. A state-sector link is available only when that overlap is observed. A separate overlap between the 2010- and 2012-base rural/urban CPI series places the July 2011–June 2012 reference-year index on the same scale.

Spatial price adjustment. The *spatial price relative* preserves cross-state and rural/urban price-level differences using the state-specific 2011–12 Tendulkar poverty lines (Planning Commission, Government of India 2013). Each state-sector poverty line is divided by the all-India rural reference of 816. A common denominator preserves the national rural-urban price difference. Direct state observations are used when available. For small states or union

territories lacking a historical index, the price table permits documented donor or all-India substitutions and records the basis for each substitution.

Household deflation and survey timing. The household price deflator is

$$P_{srt} = P_{sr}^{\text{spatial}} \times P_{srt}^{\text{temporal}},$$

where s denotes state, r rural or urban sector, and t survey time. Real MPCE is nominal MPCE divided by this deflator. For NSS rounds, I average monthly state-sector indices over the three months represented by the household's survey sub-round. For HCES, price assignment follows the household's three-month panel.

District aggregation and support rules. I attach each household's survey district to 2001 Census geography using Section A.1. Geographic assignment coverage, measured by represented person weight, ranges from 92.6 percent in 2000–01 to 100 percent in 2007–08, is about 97–98 percent in 2009–12, and is about 91.6 percent in each modern HCES round. HCES publishes consumption estimates at the national and state or union-territory levels (National Sample Survey Office 2024a). I construct the district estimates used here from household microdata and retain district sample information with every estimate. District means are person-weighted by multiplying the household survey weight by household size.

The preferred district estimates must also satisfy predeclared sample-support rules. I record the number of sampled households and primary sampling units and calculate a Kish effective sample size, which summarizes effective information under unequal survey weights. Comparisons requiring a fixed district set use the intersection of districts satisfying the relevant support rules in every compared round or outcome. Geographic assignment coverage and statistical support are therefore recorded separately.

Cross-round validation. The adjacent modern HCES rounds provide a direct validation under a nearly unchanged three-visit survey design. On preferred common support, the 2022–23 and 2023–24 district estimates have Pearson correlations of 0.859 for real mean MPCE, 0.854 for mean log real MPCE, and 0.860 for weighted median real MPCE. These correlations show consistent district variation across adjacent rounds. The older NSS rounds require the survey-definition and price adjustments described above, while the national benchmark checks verify each reconstructed nominal MPCE series before those adjustments are applied.

A.5 Additional Harmonized Outcome Panels

For the 2001–11 Census comparisons, I reconstruct migration, household human-capital, housing and asset, and worker-industry measures from published district counts (Office of the Registrar General and Census Commissioner of India 2001; Office of the Registrar General and Census Commissioner, India 2011). Eligible 2011 counts are

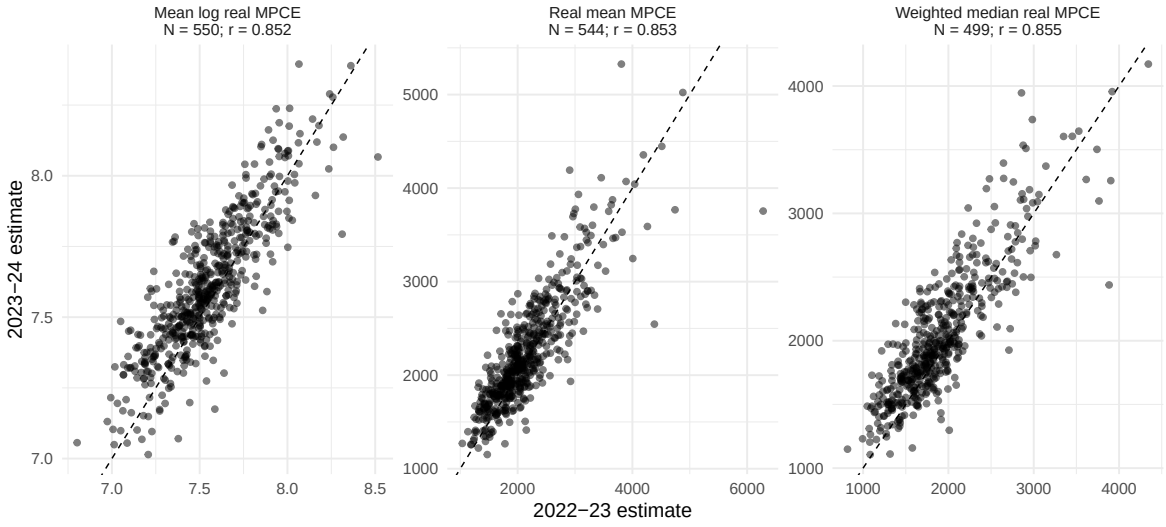


Figure 8: Agreement between district real-consumption estimates in the 2022–23 and 2023–24 HCES rounds. Each panel uses districts retained in both rounds under the consumption sample criteria; the dashed line is the 45-degree line. Sources: National Sample Survey Office, HCES 2022–23 and HCES 2023–24 (National Sample Survey Office 2024b, 2025a).

pooled onto 2001 Census geography before rates are calculated, so numerators and denominators refer to the same places in both Census years.

For the Economic Census, I use SHRUG's district products for the 2005 and 2013–14 measures that are available comparably across rounds (Asher et al. 2021). The predetermined 2005 information-technology measure requires finer industry detail, so I read the official Fifth Economic Census establishment records directly (Central Statistics Office 2008). Those records identify district, activity status, four-digit NIC-2004 activity, and total workers. I retain major-activity nonagricultural establishments and define computer-related activity as NIC-2004 Division 72. This variable is used as a 2005 baseline characteristic because the available Sixth Economic Census files do not support the same four-digit NIC-2004 definition (Central Statistics Office 2016). The 2009–10 NSS employment survey and 2017–18 PLFS provide labor-force participation and employment measures for resident workers (National Sample Survey Office 2010; National Statistical Office 2019). Economic Census employment measures workers attached to establishments. These two measurement concepts are kept separate in the analysis.

B Migration

B.1 Literature Review

Migration changes the interpretation of district outcomes. The schooling measures describe children observed in each district in 2007–08, while later consumption describes the district population many years afterward.

Population turnover changes who is represented in those later district outcomes. I therefore interpret the long-run consumption results at the district level for changing populations.

Earlier studies documented low interdistrict mobility in parts of India. Chakraborty and Bakshi (2016) report low mobility in West Bengal, Haryana, and Punjab and use district of schooling to approximate later district of residence or employment. Using earlier NSS rounds, Topalova (2005) similarly documents low decadal interdistrict mobility, especially in rural areas and for employment-related moves. Those estimates concern migration over relatively short durations. Lifetime-migrant measures accumulate moves over much longer exposure periods as respondents age (Mahapatro 2020).

Census-cohort evidence and railway passenger flows indicate substantially greater interdistrict movement after 2001 than the late-twentieth-century evidence implied (Economic Divison 2015). Recent work also links internal migration and skill acquisition to regional economic opportunities. Ghose (2026) studies India's information-technology expansion in a model where education choices and internal migration respond jointly to regional opportunities. Migration also faces substantial social and geographic frictions. Caste-based rural insurance networks can discourage migration when departure weakens access to informal insurance (Munshi and Rosenzweig 2016), and state borders sharply reduce migration between neighboring districts (Kone et al. 2018).

My migration measures come from the 2011 Census D-series tables (Office of the Registrar General and Census Commissioner, India 2011). D-02 describes migrants by place of last residence and provides the interstate share among migrants. D-03 records reason for migration. D-07 identifies migrants who have been at the destination for at most nine years, report work or employment as the reason for migration, and previously resided elsewhere in India. I call these 0–9-year work migrants *recent work migrants*. I verify the D-07 total against the corresponding D-03 work-migrant subtotal before constructing the measures. I then pool the 2011 counts to 2001 Census districts using the reviewed district concordance described in Section A.1.

Overall migrant composition combines moves for marriage, education, household relocation, and work. The recent-work-migrant measure isolates people who migrated for employment and describes the skill composition of workers newly observed at each destination. It measures work in-migrants at the destination. It contains no corresponding measure of workers leaving that district.

B.2 Empirical Results

Table 6 reports regressions with state fixed effects of migration outcomes on linguistic distance. Each coefficient compares districts within states and conditions on the predetermined Census controls used elsewhere in the paper. Panel A reports the interstate, work-reason, and education-reason composition of all migrants. Panel B reports geographic origin and educational composition among recent work migrants, together with the prespecified Hindi-

Table 6

Linguistic Distance and Migration Composition

	(1)	(2)	(3)
Panel A: All migrants			
Linguistic distance	-0.0023	0.0060	-4.329e-04
	(0.0199)	(0.0053)	(0.0010)
Dependent variable	Interstate share among migrants	Work/employment share among migrants	Education share among migrants
Sample	All states	All states	All states
Fixed effects	State	State	State
Predetermined controls	Yes	Yes	Yes
Observations	355	355	355
Panel B: Recent work migrants			
Linguistic distance	0.0050	0.0129	0.0079
	(0.0314)	(0.0049)	(0.0064)
Dependent variable	Outside-state share among recent work migrants	Graduate/technical-degree share among recent work migrants	Graduate/technical-degree share among recent work migrants
Sample	All states	All states	Hindi-belt states
Fixed effects	State	State	State
Predetermined controls	Yes	Yes	Yes
Observations	355	355	178

Each column is a separate district regression using speaker-weighted linguistic distance from Hindi, baseline 2001 Census covariates, state fixed effects, and state-clustered standard errors. National significance stars use Holm-adjusted p-values across the eight prespecified migration outcomes, five of which appear here. The prespecified Hindi-belt restriction uses its single raw p-value. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The dependent variables are shares, so coefficients are changes in shares associated with a one-degree increase in linguistic distance. The coefficients are conditional district associations between linguistic distance and migration composition. Source: Census of India 2011 D-02, D-03, and D-07 migration tables.

belt restriction.

Linguistic distance has little association with migrants' geographic origin or stated reason for migration in these regressions. A one-degree increase in linguistic distance is associated with a 0.23 percentage-point lower interstate share among all migrants (SE 1.66 percentage points), a 0.60 percentage-point higher work-reason share (SE 0.50), and a 0.04 percentage-point lower education-reason share (SE 0.10). Among recent work migrants, the outside-state share is 0.50 percentage points higher per degree (SE 2.66). The Holm-adjusted p-value is 1.000 for each of these four associations. Separate 2001–11 population-denominated migration changes are reported descriptively outside this multiple-testing family.

The educational composition estimate is positive: districts with greater linguistic distance have a higher estimated graduate-or-technical-degree share among recent work in-migrants. Under the current state-clustered small-sample inference, however, this coefficient does not survive the prespecified Holm correction, so I treat it as suggestive rather than as a statistically precise migration result. The dependent variable is the graduate-or-technical-degree share among recent work in-migrants observed at the destination.

Within the Hindi-belt states, the corresponding estimate is smaller and imprecise. Across the migration family, the geographic origin and migration reason coefficients are also small or imprecise. These results reinforce the interpretation of later consumption as a district-level outcome for a changing population without establishing a precise migration composition channel.

C Selection into School Enrollment

The district comparisons in the main text show large social-group differences in enrollment, English-medium instruction, and private schooling within the same districts. I use the child-level enrollment model to examine selection into enrollment: whether a child appears in the enrolled-child education record at all. I treat the estimates as descriptive associations and keep this analysis separate from the district-level identification design.

I use the NSS 64th Round *Participation and Expenditure in Education* survey and restrict the child roster to ages 5–19 (National Sample Survey Office 2011). I define enrollment by whether a child in the household roster has a corresponding enrolled-child education record. The probit includes age, sex, household size, religion, social group, rural or urban residence, and distance to the nearest primary class. It also includes a household education measure constructed from the education of the male household head; when no male head is available, the construction uses a married son and then a father or father-in-law. The fitted specification also includes district averages of selected schooling costs and benefits among enrolled children. Those district measures combine local provision with the composition of enrolled children, so I retain them as adjustment variables and omit their coefficients from the reported table.

The probit uses the NSS survey design, including household weights, sampled villages or urban blocks as primary sampling units, and strata formed from state, rural/urban sector, stratum, and substratum identifiers. I report quantities on the predicted-probability scale. Continuous covariates are summarized by average slopes. Categorical covariates are summarized by average discrete comparisons against the stated reference category, averaging over the observed estimation-sample covariate distribution (Hansen 2022; Arel-Bundock et al. 2024).

Table 7: Average Marginal Effects for School Enrollment

Term	Enrolled (1 = yes)
Age (years)	-0.028***
	(0.000)
Female (ref: Male)	0.044***
	(0.003)
Household size	-0.004***
	(0.001)
Religion: Muslim (ref: Hindu)	-0.126***
	(0.007)

Table 7: Average Marginal Effects for School Enrollment *(continued)*

Term	Enrolled (1 = yes)
Religion: Christian	-0.043***
	(0.013)
Religion: Sikh	0.040
	(0.026)
Religion: Jain	-0.001
	(0.012)
Religion: Buddhist	0.047*
	(0.024)
Religion: Zoroastrian	0.101***
	(0.034)
Religion: Other	0.252***
	(0.003)
Social group: Scheduled Tribe (ref: Other)	-0.102***
	(0.008)
Social group: Scheduled Caste	-0.075***
	(0.006)
Social group: Other Backward Class	-0.029***
	(0.005)
Urban (ref: Rural)	-0.003
	(0.005)
Distance 1–2km (ref: <1km)	-0.077***
	(0.026)
Distance 2–3km	-0.001
	(0.010)
Distance 3–5km	0.004
	(0.066)
Distance > 5km	0.036
	(0.035)
Male household education: Literate, no school (ref: Illiterate)	0.097***
	(0.017)
Male household education: Literate, school < primary	0.127***
	(0.006)
Male household education: Primary	0.142***
	(0.005)
Male household education: Upper primary	0.195***
	(0.006)
Male household education: Secondary	0.212***
	(0.007)
Male household education: Higher secondary	0.247***
	(0.008)
Male household education: Postsecondary+	0.245***
	(0.009)

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Entries are average changes in predicted enrollment probability. Continuous covariates are average slopes; categorical covariates are average discrete comparisons against the stated reference category, evaluated over the observed covariate distribution. Standard errors use the NSS sampling design, including household weights, primary sampling units, and strata formed from state, rural/urban sector, stratum, and substratum identifiers. The model also controls for district averages of schooling costs and benefits among enrolled children; the table displays marginal effects for child and household covariates. Source: NSS 64th Round, 2007–08, Participation and Expenditure in Education.

Social identity is strongly associated with enrollment. Conditional predicted enrollment is 12.6 percentage points lower for Muslim children than for Hindu children and 4.3 points lower for Christian children. The Muslim

difference is consistent with evidence of persistent educational disadvantage among Indian Muslims (Borooah and Sabharwal 2021). Relative to children in the residual "Other" social-group category, predicted enrollment is 10.2 points lower for Scheduled Tribe children, 7.5 points lower for Scheduled Caste children, and 2.9 points lower for Other Backward Class children. These are conditional associations. The model does not identify causal effects of religion or caste. Social institutions and caste networks can shape schooling choices and later occupational opportunities (Munshi and Rosenzweig 2006).

Male household education has the largest ordered association in the model. Relative to an illiterate designated male household member, predicted enrollment is 9.7 percentage points higher for the literate-without-schooling category, 14.2 points higher at primary education, 19.5 points higher at upper primary, 21.2 points higher at secondary, and 24.7 points higher at higher secondary. The postsecondary comparison is 24.5 points. This measure is correlated with household resources, information, and intergenerational educational advantage. The model does not separate those channels.

Within the fitted specification, one additional year of age is associated with a 2.8 percentage-point lower predicted enrollment probability on average. Girls have a 4.4-point higher predicted probability than boys conditional on the other covariates. Each additional household member is associated with a 0.37-point lower predicted probability. These estimates describe the conditional pattern in the NSS sample.

Nearby school access also has a sizable association with enrollment. Children whose nearest primary class is reported one to two kilometers away have a predicted enrollment probability 7.7 percentage points lower than children with a primary class within one kilometer. The more distant categories have wide confidence intervals and show no consistent ordering. The clearest distance pattern is therefore the one-to-two-kilometer difference.

The fitted model includes district averages of free schooling, fee waivers, scholarships, textbooks, stationery, mid-day meals, and enrollment cost among enrolled children. These terms remain adjustment variables because each combines local schooling conditions with selection into enrollment. Their coefficients are omitted from the reported table.

The child roster contains 127,181 observations. The complete-case estimation sample contains 114,898 children, leaving 12,283 observations with at least one required covariate missing. District-average enrollment cost is missing for 5.6 percent of the roster, male household education for 4.1 percent, and distance to the nearest primary class for 0.2 percent. Observed covariates explain considerably more of missingness in district-average enrollment cost and school distance than of missing male household education: the corresponding missingness-logit pseudo-R^2 values are 0.310, 0.246, and 0.023. Complete cases therefore differ systematically from omitted observations on measured characteristics. No missing-value imputation is used.

Table 8: Missingness in Child-Enrollment Model Covariates

Variable	Missing N	Missing %	Missingness pseudo-R^2
Distance to nearest primary class	312	0.2	0.246
District-average enrollment cost	7,109	5.6	0.310
Male household education	5,203	4.1	0.023
Any model covariate	12,283	9.7	

Note: The final row reports children missing at least one covariate required by the probit. Pseudo-R^2 values come from separate logits for whether each listed covariate is missing. The final row for any required covariate summarizes overall missingness and has no separate logit. No missing-value imputation is used. Source: NSS 64th Round, 2007–08, Participation and Expenditure in Education.

The child-level estimates show that unequal educational participation appears before the choice of school medium. Religion, social group, and especially male household education are strongly associated with enrollment itself, adding an earlier participation margin to the within-district inequalities in English-medium and private schooling documented in the main text.

D Linguistic Distance as an Instrumental Variable

The main text uses linguistic distance as a measure of inherited linguistic conditions. I next evaluate it as a candidate instrument for English-medium schooling. The design requires linguistic distance to predict EMI after the preferred geographic and baseline adjustment and to affect later consumption only through the schooling channel. The first requirement is weak within states, and the exclusion restriction remains substantively demanding. I therefore use this appendix to assess first-stage relevance, alternative linguistic and schooling measures, inference robust to weak identification, and sensitivity to direct effects of linguistic distance. Weak instruments can invalidate conventional 2SLS inference even when familiar first-stage rules appear reassuring (Keane and Neal 2024).

D.1 First-Stage Relevance

Linguistic distance strongly predicts EMI in the unadjusted national comparison. On the common sample used for the first-stage comparison, the excluded-instrument F statistic is 11.73 without geographic fixed effects or controls. It falls to 3.24 with region fixed effects and the main 2001 Census controls and to 0.75 with state fixed effects and the same controls. Figure 9 shows the corresponding partial relationships after applying each adjustment, and Table 9 reports the same progression across alternative linguistic measures. The partial R^2 with state fixed effects is 0.0039, so linguistic distance accounts for less than half of one percent of the residual variation in EMI exposure among all children under the preferred adjustment.

The conventional first-stage F statistic describes how relevance changes as geographic adjustment is added. For the preferred IV specifications, I assess strength with the Montiel Olea–Pflueger effective F, which is designed for weak-instrument assessment under heteroskedasticity and clustering (Montiel Olea and Pflueger 2013). A fixed rule based on the conventional F is insufficient for this purpose because the size distortions of conventional 2SLS inference depend on the design (Keane and Neal 2024).

The attenuation identifies the geographic scale at which the linguistic-distance/EMI association is concentrated. Linguistic distance and EMI are strongly related across broad Indian geography. The association becomes much smaller when districts are compared within the same state. This pattern is consistent with the state-level organization of English-medium schooling documented in the main text, especially in government schools. The remaining within-state variation in linguistic distance therefore provides little predictive content for EMI exposure.

D.2 Spatial Dependence

EMI exposure, linguistic distance, consumption growth, and changes in consumption inequality are geographically concentrated, so I test the preferred first- and second-stage residuals for positive spatial autocorrelation. I

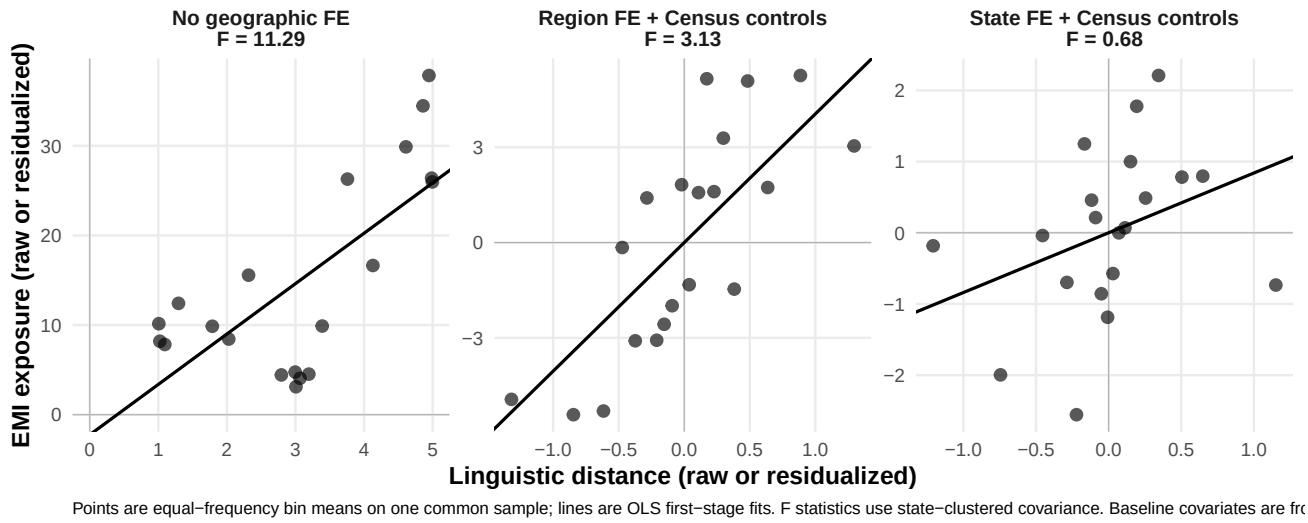


Figure 9: Linguistic distance and EMI exposure among all children under no geographic fixed effects, region fixed effects, and state fixed effects. Points are equal-frequency bin means on a common sample; lines are OLS first-stage fits. The region and state panels residualize both axes using the corresponding fixed effects and baseline 2001 Census covariates. Panel labels report state-clustered first-stage F statistics. Sources: Census of India 2001 and NSS 64th Round, 2007–08 (Office of the Registrar General and Census Commissioner of India 2001; National Sample Survey Office 2011; Asher et al. 2021).

construct row-standardized contiguity weights from the 2001 Census district polygons and calculate Moran's I, a global measure of whether neighboring districts have similar values (Anselin 2001). Under rook contiguity, English-medium exposure has $I = 0.661$, linguistic distance has $I = 0.886$, real consumption growth has $I = 0.307$, and the change in the Gini coefficient has $I = 0.143$; each test rejects the null of no positive spatial autocorrelation at $p < 0.001$.

For the preferred consumption specification, the second-stage residuals have Moran's $I = 0.00264$ (permutation $p = 0.426$), and the first-stage residuals have $I = -0.0103$ (permutation $p = 0.617$). Neither test rejects the null of no positive residual spatial autocorrelation. Queen contiguity gives the same result: the second-stage residual test has permutation $p = 0.417$ and the first-stage residual test has permutation $p = 0.63$. Lakshadweep, South Andaman, and Nicobars have no contiguous district neighbor and enter with zero spatial lag. The raw variables are strongly spatially clustered, while the preferred residual tests provide no evidence of positive spatial autocorrelation under either contiguity definition.

D.3 Alternative Measures and Historical Constructions

The same weak within-state relevance appears across alternative linguistic measures. Table 9 compares the preferred speaker-weighted measure with scalar alternatives based on the largest languages, genealogical dis-

Table 9

First-Stage Relevance Across Linguistic Measures

Linguistic Measure	No Geographic FE F	Region FE F	State FE F	State Partial R^2	N
Speaker-weighted distance from Hindi	11.29	2.01	0.59	0.004	573
Top-three-language distance from Hindi	15.52	2.79	1.80	0.012	573
Genealogical distance from Hindi	8.11	0.23	0.28	0.001	573
Lexical noncognacy with Hindi	4.89	0.04	0.15	0.000	573
Share speaking languages distant from Hindi	3.79	0.35	0.54	0.004	573
Language-distance composition (5 instruments)	2.85	3.56	0.97	0.041	573
Historical 1991 speaker-weighted distance	9.36	0.22	0.04	0.000	89

Note: The first three columns report the state-clustered joint excluded-instrument F statistic for the unadjusted specification, six-region fixed effects plus baseline 2001 Census covariates, and state fixed effects plus the same covariates. The final relevance column is the partial R^2 with state fixed effects. The historical row uses districts available for the validated 1991 reconstruction and is estimated separately from the modern district sample. These conventional first-stage statistics describe relevance across geographic adjustments. The consumption IV specifications use the Montiel Olea–Pflueger effective F statistic for weak-identification assessment. Sources: Census of India 1991 Language Atlas; Census of India 2001; NSS 64th Round, 2007–08.

tance, lexical noncognacy, and the population share speaking languages distant from Hindi. It also reports a five-component language-distance composition and the independently reconstructed 1991 measure. For each construction, I report the excluded-instrument F statistic without geographic fixed effects, with region fixed effects, and with state fixed effects. The final column gives the partial R^2 with state fixed effects.

Each scalar alternative loses most of its first-stage relevance under state fixed effects. The top-three measure falls from an F statistic of 16.30 to 1.96. The genealogical and lexical measures fall from 10.43 and 5.29 to 0.28 and 0.15, and the distant-language share falls from 4.08 to 0.59. The five-component specification retains more within-state predictive content than the scalar measures: its joint F with state fixed effects is 3.73 and its partial R^2 is 0.041. These values remain small for an overidentified IV specification with five excluded instruments.

The independently reconstructed 1991 measure is even weaker under geographic adjustment. On its validated common support, the 1991 speaker-weighted distance has an F statistic of 9.98 without geographic fixed effects, 0.31 with region fixed effects and Census controls, and 0.03 with state fixed effects and the same controls. Measuring the linguistic environment in 1991 gives it clear temporal precedence over the 2007–08 schooling measure, while exclusion still depends on assumptions beyond timing, pretreatment balance, and negative-control checks (Danieli et al. 2026).

The NSS–DISE comparison shows the same weak relevance after state fixed effects with an independently re-

Table 10

Inference with Weak Instruments for Long-Run Consumption Change

	Real log consumption change, 2004-05 to 2022-23	Real log consumption change, 2004-05 to 2023-24
EMI exposure	0.117	0.137
	(0.182)	(0.281)
MOP effective F	0.69	0.41
AR p-value at zero	0.029	0.081
AR 95% accepted set	$[-0.193, -0.030]^{\dagger} \cup [0.009, 0.193]^{\dagger}$	$[-0.186, 0.186]^{\dagger}$
Direct effect threshold for $\beta = 0$ (share of |reduced form|)	16.6%	0.0%
Observations	524	522

The table reports conventional 2SLS coefficients with state-clustered standard errors. Instrument strength is summarized by the Montiel Olea–Pflueger effective F statistic, and Anderson–Rubin tests provide inference robust to weak instruments. The 95% AR accepted sets are displayed as unions of closed intervals. †A component reaches an evaluated search boundary; the displayed outer endpoint is the numerical search limit. The direct effect threshold is the smallest permitted direct effect, expressed as a share of the absolute reduced form, for which the AR set includes $\beta = 0$. Sources: Census of India 2001; NSS 61st Round, 2004–05; NSS 64th Round, 2007–08; HCES 2022–23; HCES 2023–24.

constructed schooling measure. The two EMI measures correlate at about 0.896 in district levels and 0.607 after removing state means (Figure 7). EMI measurement therefore retains substantial within-state agreement across the household and school data even though linguistic distance predicts that variation weakly. Alternative language classifications and distance constructions lead to the same conclusion after state fixed effects.

The richer language-share specifications also remain weak under state fixed effects. All three falsification-adaptive sets with state fixed effects include zero. A falsification-adaptive set reports structural values that remain admissible under the smallest relaxations needed to reconcile the overidentifying restrictions with the data (Masten and Poirier 2021). These sets therefore leave the sign undetermined.

D.4 Inference Robust to Weak Identification

The weak first stage changes the appropriate inference for the second stage. I report conventional 2SLS coefficients for completeness and base inference on methods that remain valid under weak identification. The preferred assessment uses the Montiel Olea–Pflueger effective F statistic and Anderson–Rubin (AR) tests. The AR test evaluates candidate structural coefficients without requiring a strong first stage (Keane and Neal 2024).

Table 10 reports the two long-run change specifications used in the main identification discussion. The table places the conventional 2SLS coefficients beside the weak-identification statistics and omits conventional significance stars.

The table reports the conventional 2SLS estimates and state-clustered standard errors alongside the Montiel

Olea–Pflueger effective F statistics. I use the weak-identification-robust results, rather than conventional 2SLS t tests, for structural inference.

The AR rows make the inferential consequence explicit. The table reports the point-null p-value at $\beta = 0$ and the 95% accepted set for each endpoint. A regular inversion grid discovers the accepted-set topology, while interior acceptance boundaries are numerically refined; components that still reach the finite search range are marked with a dagger rather than interpreted as finite confidence bounds. The table therefore reports the topology returned by the test rather than imposing a finite-interval or common-sign interpretation.

The robustness families reach the same conclusion. They vary the linguistic measure, EMI treatment, welfare outcome, controls, parameterization, and historical adjustment. Together they contain 408 model specifications across seven families, and none satisfies the effective-F strength criterion. I retain the model-level results in the replication outputs and summarize the family-level evidence here.

I also evaluate sensitivity to violations of the exclusion restriction. I use a bounded sensitivity analysis for direct effects in the spirit of plausibly exogenous instruments, allowing linguistic distance to have a direct effect on the outcome and asking how large the permitted effect must be before $\beta = 0$ enters the AR confidence set (Conley et al. 2012). Table 10 reports that threshold as a share of the absolute reduced form for each endpoint. A zero threshold means that $\beta = 0$ already belongs to the exact-exclusion AR accepted set. These are sensitivity thresholds, not estimates of the direct effect of linguistic distance, and the first stage remains weak in both specifications.

Accordingly, linguistic distance remains a descriptive measure of inherited linguistic conditions in the main analysis. Its within-state relevance for EMI is extremely weak, and its associations with several other local characteristics make exact exclusion substantively demanding. The schooling–welfare estimates in the main text are therefore descriptive district associations.

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